



Nearly Toroidal, Periodic and Quasi-periodic Motions of Fluid Particles Driven by the Gavrilov Solutions of the Euler Equations

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Communicated by A. Vasseur

Abstract. We consider the smooth, compactly supported solutions of the steady 3D Euler equations of incompressible fluids constructed by Gavrilov (Geom Funct Anal (GAFA) 29(1):190–197, 2019), and we study the corresponding fluid particle dynamics. This is an ODE analysis, which contributes to the description of Gavrilov’s vector field.

Mathematics Subject Classification. 35Q31, 37N10, 76B03.

Keywords. 3D Euler equations, Gavrilov solutions, Fluid particle dynamics, Periodic motions, Quasi-periodic motions, Frequency function.

Contents

1. Introduction and Main Result
 - 1.1. The Gavrilov Solutions of the Steady Euler Equations
 - 1.2. Main Result: Description of the Fluid Particle Dynamics
 - 1.3. Related Literature
2. Dual Role of the Parameter R in Gavrilov’s Solutions
3. Conjugation to a Linear Flow on $\mathbb{T}^2$
 - 3.1. Cylindrical Coordinates
 - 3.2. Elimination of the Factor $1/\rho$ and Canonical Hamiltonian Structure
 - 3.3. Symplectic Polar Coordinates in the Radial–Vertical Plane
 - 3.4. Level Sets
 - 3.5. Integration of the System in the Radial–Vertical Plane
 - 3.6. Rotation Period in the Radial–Vertical Plane
 - 3.7. Angle–Action Variables in the Radial–Vertical Plane
 - 3.8. Reduction to a Constant Rotation in the Tangential Direction
 - 3.9. Ratio of the Two Rotation Periods
 - 3.10. Motion of the Fluid Particles
 - 3.11. The Pressure in Terms of the Action
 - 3.12. Volume of the Region Enclosed by a Level Set
4. Taylor Expansions and Transversality
 - 4.1. Expansion of α
 - 4.2. Expansion of α_2
 - 4.3. Expansion of $\gamma_c(\vartheta)$
 - 4.4. Expansion of the Average of $\gamma_c(\vartheta)$ and $\partial_c \gamma_c(\vartheta)$
 - 4.5. Expansion of $J(c)$
 - 4.6. Expansion of $h(I)$ and $\Omega_1(I)$
 - 4.7. Expansion of the Frequency Ratio $\Omega_2(I)/\Omega_1(I)$

4.8. Expansion of $\Phi(\sigma, \beta, I)$

4.9. Smallness Conditions

References

1. Introduction and Main Result

In the remarkable paper [10], Gavrilo proved the existence of a nontrivial solution of C^∞ class, with compact support, of the steady Euler equations of incompressible fluids in $\mathbb{R}^3$. The result in [10] is important and surprising, because previously, on the basis of some negative partial results, it was conjectured that compactly supported, nontrivial, smooth solutions of the 3D steady Euler equations cannot exist: see the clear explanation at the beginning of [5] and the general discussion about existence of compactly supported smooth solutions in [13]. In addition, another reason of interest for the fruitful construction of [10] is that recently it has been used as a building block to produce other interesting solutions, both stationary and time-dependent, of the Euler equations of fluid dynamics, see Sect. 1.3 below.

Now suppose that a fluid moves according to the Gavrilo solution of the Euler equations, that is, suppose that the fluid particles are driven by Gavrilo's velocity vector field. Which movement of the fluid do we observe? Of course the particles outside the support of the vector field do not move at all, but how do they move in the region where the field is nonzero?

In the present paper we deal with this question. It turns out that every fluid particle travels along a trajectory that lies on a nearly toroidal surface, which is a level set of the pressure. The motion of every fluid particle is periodic or quasi-periodic in time; we prove that, in fact, there are both periodic and quasi-periodic motions, and the value of the pressure determines whether the trajectories on its level set are all periodic or all quasi-periodic.

In fact, the system of differential equations in $\mathbb{R}^3$ describing the motion of the fluid particles turns out to be integrable, as it can be transformed into the system of a Hamiltonian system of one degree of freedom and a third equation that can be directly solved by integration. We write the Hamiltonian system in angle-action coordinates (σ, I) , and prove that there exists a change of variables on a neighborhood of the support of Gavrilo's vector field such that the equations of motion in the new coordinates (σ, β, I) becomes

$$\dot{\sigma} = \Omega_1(I), \quad \dot{\beta} = \Omega_2(I), \quad \dot{I} = 0,$$

where σ and β are angle variables rotating with constant angular velocities $\Omega_1(I)$, $\Omega_2(I)$, and I is a constant action variable, which is, in fact, a reparametrization of the pressure. The full statement is in Theorem 1.1.

1.1. The Gavrilo Solutions of the Steady Euler Equations

In the main part of the construction in [10], given any $R > 0$, the circle

$$\mathcal{C} := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = R^2, z = 0\}$$

in $\mathbb{R}^3$ is considered, and, in an open neighborhood $\mathcal{N}$ of $\mathcal{C}$, two functions U, P are defined, $U : \mathcal{N} \rightarrow \mathbb{R}^3$ and $P : \mathcal{N} \rightarrow \mathbb{R}$, both in $C^\infty(\mathcal{N} \setminus \mathcal{C})$, solving the steady Euler equations

$$U \cdot \nabla U + \nabla P = 0, \quad \operatorname{div} U = 0 \tag{1.1}$$

in $\mathcal{N} \setminus \mathcal{C}$, together with the fundamental “localizability condition”

$$U \cdot \nabla P = 0. \tag{1.2}$$

As a final step of the proof, the functions U, P are multiplied by smooth cut-off functions to obtain $C^\infty(\mathbb{R}^3)$ functions $\tilde{U}, \tilde{P}$, where $\tilde{U}$ and $\nabla \tilde{P}$ have compact support contained in $\mathcal{N} \setminus \mathcal{C}$, solving (1.1) (and also (1.2)) in $\mathbb{R}^3$.

Let us be more precise. Denote $\rho := \sqrt{x^2 + y^2}$. For $\delta \in (0, R)$, let

$$\mathcal{N} = \{(x, y, z) \in \mathbb{R}^3 : (\rho - R)^2 + z^2 < \delta^2\}. \quad (1.3)$$

In $\mathcal{N}$, the solution (U, P) of [10] is given by

$$U(x, y, z) = u_\rho(\rho, z)e_\rho(x, y) + u_\varphi(\rho, z)e_\varphi(x, y) + u_z(\rho, z)e_z, \quad P(x, y, z) = p(\rho, z), \quad (1.4)$$

where

$$\begin{aligned} e_\rho(x, y) &= \frac{1}{\rho}(x, y, 0), & e_\varphi(x, y) &= \frac{1}{\rho}(-y, x, 0), & e_z &= (0, 0, 1), \\ u_\rho(\rho, z) &= \frac{\partial_z p(\rho, z)}{\rho}, & u_\varphi(\rho, z) &= \frac{b(\rho, z)}{\rho}, & u_z(\rho, z) &= -\frac{\partial_\rho p(\rho, z)}{\rho}, \\ b(\rho, z) &= \frac{R^3}{4}\sqrt{H(a(\rho, z))}, & p(\rho, z) &= \frac{R^4}{4}a(\rho, z), & a(\rho, z) &= \alpha\left(\frac{\rho}{R}, \frac{z}{R}\right), \end{aligned} \quad (1.5)$$

and α, H are functions defined in [10] in terms of solutions of certain differential equations; $H(0) = 0$, and H is analytic in a neighborhood of 0; α has a strict local minimum at $(1, 0)$, with $\alpha(1, 0) = 0$, and it is analytic in a neighborhood of $(1, 0)$. Hence α and $H \circ \alpha$ are both well-defined and analytic in a disc of $\mathbb{R}^2$ of center $(1, 0)$ and radius r_0 , for some universal constant $r_0 > 0$ (where “universal” means that r_0 does not depend on anything). If δ in (1.3) satisfies

$$\delta \leq r_0 R,$$

then $a(\rho, z)$ and $H(a(\rho, z))$, where $\rho = \sqrt{x^2 + y^2}$, are well-defined and analytic in $\mathcal{N}$ (note that, in $\mathcal{N}$, one has $0 < R - \delta < \rho < R + \delta$; in particular, ρ is bounded away from zero). Also, $b(\rho, z)$ is well-defined and continuous in $\mathcal{N}$, and (because of the square root $\sqrt{H}$) it is analytic in $\mathcal{N} \setminus \mathcal{C}$. Hence P is analytic in $\mathcal{N}$, while U is continuous in $\mathcal{N}$ and analytic in $\mathcal{N} \setminus \mathcal{C}$.

The solution $(\tilde{U}, \tilde{P})$ in [10] is defined in $\mathcal{N}$ as

$$\tilde{U}(x, y, z) = \omega(P(x, y, z))U(x, y, z), \quad \tilde{P}(x, y, z) = W(P(x, y, z)), \quad (1.6)$$

where $\omega : \mathbb{R} \rightarrow \mathbb{R}$ is any C^∞ function vanishing outside the interval $[\varepsilon, 2\varepsilon]$, with $\varepsilon > 0$ small enough, and $W : \mathbb{R} \rightarrow \mathbb{R}$ is a primitive of ω^2 ; for example,

$$W(s) = \int_0^s \omega^2(\sigma) d\sigma. \quad (1.7)$$

Then $(\tilde{U}, \tilde{P})$ is extended to $\mathbb{R}^3$ by defining $(\tilde{U}, \tilde{P}) = (0, W(2\varepsilon))$ in $\mathbb{R}^3 \setminus \mathcal{N}$.

Note that $\tilde{U}$ and $\nabla \tilde{P}$ can be nonzero only in the set

$$\mathcal{S} := \{(x, y, z) \in \mathcal{N} : \varepsilon < P(x, y, z) < 2\varepsilon\}, \quad (1.8)$$

and, if $\omega(s)$ is nonzero at some $s \in (\varepsilon, 2\varepsilon)$, then both $\tilde{U}$ and $\nabla \tilde{P}$ are actually nonzero at the corresponding points in $\mathcal{S}$. Moreover, $P = 0$ on the circle $\mathcal{C}$, and, for ε small enough, $P > 3\varepsilon$ at all points of $\mathcal{N}$ sufficiently far from $\mathcal{C}$; more precisely, to fix the details of the construction, we introduce a parameter $\tau > 0$ and assume that

$$P(x, y, z) > \tau \quad \forall (x, y, z) \in \mathcal{N} \setminus \mathcal{N}' \quad (1.9)$$

and $\tau \geq 3\varepsilon$, where

$$\mathcal{N}' := \{(x, y, z) \in \mathbb{R}^3 : (\rho - R)^2 + z^2 < (\delta/4)^2\}.$$

Thus, the closure of $\mathcal{S}$ is contained in the open set

$$\mathcal{S}^* := \{(x, y, z) \in \mathcal{N} : 0 < P(x, y, z) < \tau\}, \quad (1.10)$$

and $\mathcal{S}^* \subseteq (\mathcal{N}' \setminus \mathcal{C}) \subseteq (\mathcal{N} \setminus \mathcal{C})$.

1.2. Main Result: Description of the Fluid Particle Dynamics

A preliminary, basic observation regarding the solutions of Sect. 1.1 is that any such solution with a given $R > 0$ can be obtained, by rescaling, from another one having $R = 1$. Even more, we show that Gavrilov's solutions are a 1-parameter subset of a larger 2-parameter family of solutions, where R plays a dual role related to two different scaling invariances of the Euler equations, both preserving the localizability condition (1.2). These basic observations are in Sect. 2 (see Lemma 2.1). Thanks to these properties, we study the motion of the fluid particles in the normalized case $R = 1$; the motion for any other $R > 0$ is immediately obtained by rescaling the amplitude and the time variable, as explained in Lemma 2.2.

To study the motion of the fluid particles driven by the Gavrilov vector field $\tilde{U}$ means to study the solutions $\mathbb{R} \rightarrow \mathbb{R}^3$, $t \mapsto (x(t), y(t), z(t))$ of the system

$$(\dot{x}(t), \dot{y}(t), \dot{z}(t)) = \tilde{U}(x(t), y(t), z(t)), \quad (1.11)$$

which is an autonomous ODE in $\mathbb{R}^3$. The dot above a function denotes its time derivative. Before dealing with system (1.11), we recall some definitions about quasi-periodic functions.

A vector $\Omega = (\Omega_1, \dots, \Omega_n) \in \mathbb{R}^n$, $n \geq 1$, is said to be *rationally independent* if $\Omega \cdot k = \Omega_1 k_1 + \dots + \Omega_n k_n$ is nonzero for all integer vectors $k = (k_1, \dots, k_n) \in \mathbb{Z}^n \setminus \{0\}$.

Given a set X , a function $v : \mathbb{R} \rightarrow X$, $t \mapsto v(t)$ is said to be *quasi-periodic with frequency vector* $\Omega = (\Omega_1, \dots, \Omega_n) \in \mathbb{R}^n$, $n \geq 2$, if Ω is rationally independent and there exists a function $w : \mathbb{R}^n \rightarrow X$, 2π -periodic in each real variable, such that $v(t) = w(\Omega_1 t, \dots, \Omega_n t)$ for all $t \in \mathbb{R}$. Moreover, to ensure that the number n is not higher than necessary, we add the condition that there does not exist any vector $\tilde{\Omega} = (\tilde{\Omega}_1, \dots, \tilde{\Omega}_m) \in \mathbb{R}^m$, with $m < n$, and any function $\tilde{w} : \mathbb{R}^m \rightarrow X$, 2π -periodic in each real variable, such that $v(t) = \tilde{w}(\tilde{\Omega}_1 t, \dots, \tilde{\Omega}_m t)$ for all $t \in \mathbb{R}$.

For example, for $X = \mathbb{R}$, $n = 3$, and $\Omega = (\Omega_1, \Omega_2, \Omega_3) \in \mathbb{R}^3$ a rationally independent vector, if $w(\vartheta_1, \vartheta_2, \vartheta_3) = \cos(\vartheta_1 + \vartheta_2) + \cos(\vartheta_3)$, then $n = 3$ is not minimal, as it can be reduced to $n = 2$ by taking $\tilde{w}(\vartheta_1, \vartheta_2) = \cos(\vartheta_1) + \cos(\vartheta_2)$, $\tilde{\Omega} = (\tilde{\Omega}_1, \tilde{\Omega}_2) \in \mathbb{R}^2$, with $\tilde{\Omega}_1 = \Omega_1 + \Omega_2$ and $\tilde{\Omega}_2 = \Omega_3$, while $n = 2$ cannot be further reduced. Hence the function $v(t) = \tilde{w}(\tilde{\Omega}_1 t, \tilde{\Omega}_2 t) = \cos(\tilde{\Omega}_1 t) + \cos(\tilde{\Omega}_2 t)$ is quasi-periodic with frequency vector $\tilde{\Omega} \in \mathbb{R}^2$.

For $n = 2$, a vector $\Omega = (\Omega_1, \Omega_2) \in \mathbb{R}^2$ is rationally independent if and only if Ω_1 is nonzero and the ratio Ω_2/Ω_1 is irrational. Hence a function $v(t) = w(\Omega_1 t, \Omega_2 t)$ is quasi-periodic with frequency vector $\Omega = (\Omega_1, \Omega_2)$ if $w(\vartheta_1, \vartheta_2)$ is 2π -periodic in ϑ_1 and in ϑ_2 , Ω_1 is nonzero, Ω_2/Ω_1 is irrational, and $v(t)$ is not a periodic function.

The main result of this paper is the following description of Gavrilov's fluid particle dynamics.

Theorem 1.1. *There exist universal positive constants $\delta_0, \tau_0, \varepsilon_0, I^*$ with the following properties. Let*

$$\mathcal{C}, \delta, \mathcal{N}, U, P, \varepsilon, \omega, W, \tilde{U}, \tilde{P}, \mathcal{S}, \tau, \mathcal{N}', \mathcal{S}^*$$

be the sets, constants, and functions defined in Sect. 1.1 for $R = 1$, with $\delta = \delta_0$, $\tau = \tau_0$, and $0 < \varepsilon \leq \varepsilon_0$. In particular, ω is any smooth cut-off function of the Gavrilov construction, so that the composition $\omega \circ P$, and therefore the functions $\tilde{U}$ and $\nabla \tilde{P}$, vanish outside the set $\mathcal{S}$; moreover, the set $\mathcal{S}^$ is an open neighborhood of the closure of $\mathcal{S}$, and it is independent of $\omega, \mathcal{S}, \varepsilon$.*

(i) There exists an analytic diffeomorphism

$$\Phi : \mathbb{T} \times \mathbb{T} \times (0, I^*) \rightarrow \mathcal{S}^*,$$

where $\mathbb{T} := \mathbb{R}/2\pi\mathbb{Z}$, such that the change of variable $(x, y, z) = \Phi(\sigma, \beta, I)$ transforms system (1.11) into a system of the form

$$\dot{\sigma} = \Omega_1(I), \quad \dot{\beta} = \Omega_2(I), \quad \dot{I} = 0. \quad (1.12)$$

As a consequence, the solution $(x(t), y(t), z(t))$ of the Cauchy problem (1.11) with initial datum

$$(x(0), y(0), z(0)) = (x_0, y_0, z_0) = \Phi(\sigma_0, \beta_0, I_0) \in \mathcal{S}^* \quad (1.13)$$

is the function

$$(x(t), y(t), z(t)) = \Phi(\sigma(t), \beta(t), I(t)), \quad (1.14)$$

defined for all $t \in \mathbb{R}$, where

$$\sigma(t) = \sigma_0 + \Omega_1(I_0)t, \quad \beta(t) = \beta_0 + \Omega_2(I_0)t, \quad I(t) = I_0. \quad (1.15)$$

The angle variables $\sigma(t), \beta(t) \in \mathbb{T}$ rotate with constant (possibly zero) angular frequencies $\Omega_1(I_0), \Omega_2(I_0)$ respectively, and the variable $I(t) = I_0$ is constant in time.

(ii) The first and third equations of the transformed system (1.12) form a Hamiltonian system

$$\dot{\sigma} = \partial_I \mathcal{H}(\sigma, I), \quad \dot{I} = -\partial_\sigma \mathcal{H}(\sigma, I) \quad (1.16)$$

with Hamiltonian $\mathcal{H}(\sigma, I) = \mathcal{H}(I)$ independent of the angle variable σ ; hence (1.16) is a Hamiltonian system in angle-action variables.

(iii) The frequency $\Omega_1(I)$ is given by the product

$$\Omega_1(I) = \omega(\mathcal{K}(I)) \mathcal{K}'(I)$$

where ω is the C^∞ cut-off function in the assumptions, and $\mathcal{K}(I)$ is the restriction to the interval $(0, I^*)$ of a function defined and analytic in the interval $(-I^*, I^*)$, with Taylor expansion

$$\mathcal{K}(I) = I + \frac{1065}{1024} I^3 + O(I^4)$$

around $I = 0$, and strictly increasing in $(-I^*, I^*)$. The frequency $\Omega_2(I)$ is given by the product

$$\Omega_2(I) = \sqrt{I} \mathcal{R}(I) \Omega_1(I)$$

where $\mathcal{R}(I)$ is the restriction to the interval $(0, I^*)$ of a function defined and analytic in the interval $(-I^*, I^*)$, with Taylor expansion

$$\mathcal{R}(I) = 1 + \frac{7}{4} I + O(I^2)$$

around $I = 0$. If $\omega(\mathcal{K}(I)) \neq 0$, then both $\Omega_1(I)$ and $\Omega_2(I)$ are nonzero, with ratio

$$\frac{\Omega_2(I)}{\Omega_1(I)} = \sqrt{I} \mathcal{R}(I).$$

The quantity $\sqrt{I} \mathcal{R}(I)$ is well-defined also if $\omega(\mathcal{K}(I)) = 0$, because $\mathcal{R}(I)$ is independent of ω . The function $I \mapsto \sqrt{I} \mathcal{R}(I)$ is strictly increasing and analytic in $(0, I^*)$. Therefore it is rational for infinitely many values of I , and irrational for infinitely many other values of I . More precisely, the set $\{I \in (0, I^*) : \sqrt{I} \mathcal{R}(I) \in \mathbb{Q}\}$ is a countable set, while the set $\{I \in (0, I^*) : \sqrt{I} \mathcal{R}(I) \notin \mathbb{Q}\}$ has full Lebesgue measure.

(iv) For $\omega(\mathcal{K}(I_0)) \neq 0$, the solution (1.14) of the Cauchy problem (1.11), (1.13) is periodic in time if $\sqrt{I_0} \mathcal{R}(I_0)$ is rational, and it is quasi-periodic in time with frequency vector $(\Omega_1(I_0), \Omega_2(I_0))$ if $\sqrt{I_0} \mathcal{R}(I_0)$ is irrational. For $\omega(\mathcal{K}(I_0)) = 0$, the solution (1.14) of the Cauchy problem (1.11), (1.13) is constant in time, and it coincides with the initial datum.

(v) The map $\Phi(\sigma, \beta, I)$ admits a converging expansion in powers of $\sqrt{I}$ around $I = 0$; more precisely, there exists a map $\Psi(\sigma, \beta, \mu)$, defined and analytic in $\mathbb{T}^2 \times (-\mu_0, \mu_0)$, where $\mu_0 = \sqrt{I^*}$, such that $\Phi(\sigma, \beta, I) = \Psi(\sigma, \beta, \sqrt{I})$ for all $(\sigma, \beta, I) \in \mathbb{T}^2 \times (0, I^*)$. The map $\Phi(\sigma, \beta, I)$ has the form

$$\Phi(\sigma, \beta, I) = \begin{pmatrix} \varrho(\sigma, I) \cos(\beta + \eta(\sigma, I)) \\ \varrho(\sigma, I) \sin(\beta + \eta(\sigma, I)) \\ \zeta(\sigma, I) \end{pmatrix} \quad (1.17)$$

where the functions $\varrho(\sigma, I), \eta(\sigma, I), \zeta(\sigma, I)$ have expansion

$$\varrho(\sigma, I) = 1 + \sqrt{2I} \sin \sigma + O(I), \quad \eta(\sigma, I) = O(I), \quad \zeta(\sigma, I) = \sqrt{2I} \cos \sigma + O(I).$$

The map Φ can be continuously extended to $I = 0$, with $\Phi(\sigma, \beta, 0) = (\cos \beta, \sin \beta, 0)$, so that $\Phi(\mathbb{T}^2 \times \{0\}) = \mathcal{C}$.

(vi) The action variable I and the pressure P in (1.4) are related by the identity

$$P(\Phi(\sigma, \beta, I)) = \mathcal{K}(I) \quad \forall (\sigma, \beta, I) \in \mathbb{T}^2 \times (0, I^*).$$

The pressure level and the action are in bijective correspondence; thus, the action is a reparametrization of the pressure P . The frequencies $\Omega_1(I), \Omega_2(I)$ could also be expressed in terms of the pressure P . The pressure $\tilde{P}$ in (1.6) satisfies

$$\tilde{P}(\Phi(\sigma, \beta, I)) = W(\mathcal{K}(I)),$$

where W is defined in (1.7).

(vii) The orbit $\{(x(t), y(t), z(t)) : t \in \mathbb{R}\}$ of the solution (1.14) lies in the level set

$$\mathcal{T}_\ell = \{(x, y, z) \in \mathcal{N} : P(x, y, z) = \ell\} \subset \mathcal{S}^*$$

of the pressure P , where the value $\ell = P(x_0, y_0, z_0) \in (0, \tau_0)$ is determined by the initial datum (x_0, y_0, z_0) in (1.13). The level set $\mathcal{T}_\ell$ has a nearly toroidal shape, that is, $\mathcal{T}_\ell$ is close to an axisymmetric torus with circular cross section, because $P(x, y, z)$ is given by (1.4), (1.5), and

$$\alpha(\rho, z) = 2(\rho - 1)^2 + 2z^2 + O(|\rho - 1| + |z|)^3$$

around $(\rho, z) = (1, 0)$. In fact, the surface $\mathcal{T}_\ell$ is the diffeomorphic image

$$\mathcal{T}_\ell = \Phi(\mathbb{T}^2 \times \{I\})$$

of the 2-dimensional torus $\mathbb{T}^2 \times \{I\} = \{(\sigma, \beta, I) : (\sigma, \beta) \in \mathbb{T}^2\}$, where I is determined by the identity

$$\ell = \mathcal{K}(I). \quad (1.18)$$

The pressure level $\ell = P(x_0, y_0, z_0)$ of a point $(x_0, y_0, z_0) \in \mathcal{S}^*$ determines whether the solution (1.14), (1.15) of the Cauchy problem (1.11) with initial datum (1.13) on the surface $\mathcal{T}_\ell$ is constant, periodic or quasi-periodic: if $\omega(\ell) = 0$, then the solution is constant in time; if $\omega(\ell) \neq 0$ and $\Omega_2(I)/\Omega_1(I) \in \mathbb{Q}$, where I and ℓ are related by (1.18), then the solution is periodic in time; if $\omega(\ell) \neq 0$ and $\Omega_2(I)/\Omega_1(I) \notin \mathbb{Q}$, then the solution is quasi-periodic in time, with frequency vector $(\Omega_1(I), \Omega_2(I))$. Different solutions of system (1.11) lying on the same surface $\mathcal{T}_\ell$, i.e., having the same level of pressure P , share the same frequencies $\Omega_1(I), \Omega_2(I)$.

If $\omega(\ell) \neq 0$, or if ℓ is an isolated zero of ω , then the function W in (1.7) is strictly increasing in a neighborhood of ℓ , and the surface $\mathcal{T}_\ell$ is also the level set $\{(x, y, z) \in \mathcal{N} : \tilde{P}(x, y, z) = W(\ell)\}$ of the pressure $\tilde{P}$ in (1.6).

(viii) The Jacobian determinant of Φ is

$$\det D\Phi(\sigma, \beta, I) = 1 \quad \forall (\sigma, \beta, I) \in \mathbb{T}^2 \times (0, I^*).$$

Hence the map Φ is volume-preserving (where the volume of any subset of $\mathbb{T}^2 \times (0, I^*)$ is defined as the 3-dimensional Lebesgue measure on the box $[0, 2\pi]^2 \times (0, I^*)$; in other words, the integral of any periodic function is its Lebesgue integral over one period).

The relation (1.18) between the pressure and the action has the following geometrical property: for all $\ell \in (0, \tau_0)$, the bounded region E_ℓ enclosed by the surface $\mathcal{T}_\ell$, namely the open set

$$E_\ell = \{(x, y, z) \in \mathcal{N} : P(x, y, z) < \ell\}, \quad (1.19)$$

satisfies

$$E_\ell \setminus \mathcal{C} = \Phi(\mathbb{T}^2 \times (0, I)),$$

where I and ℓ are related by (1.18), and

$$\text{vol}(E_\ell) = (2\pi)^2 I,$$

where vol is the Lebesgue measure in $\mathbb{R}^3$.

(ix) The constants $\delta_0, \tau_0, \varepsilon_0, I^*$, the set $\mathcal{S}^*$, and the functions $\Phi, \mathcal{K}, \mathcal{R}, \varrho, \eta, \zeta$ are all universal, in the sense that they do not depend on any parameter; in particular, they are independent of $\omega, \varepsilon, \mathcal{S}$.

Remark 1.2 (*Support of the cut-off function*). Theorem 1.1 is stated for cut-off functions ω supported in $[\varepsilon, 2\varepsilon]$, like those in [10]; however, Theorem 1.1, as well as the result of [10], also holds for ω supported in any interval $[\varepsilon_1, \varepsilon_2]$ with $0 < \varepsilon_1 < \varepsilon_2 \leq \varepsilon_0$, without changing anything in the proof.

Remark 1.3 (*3-dimensional invariant set of a Hamiltonian system with 2 degrees of freedom*). By (1.16), system (1.12) is equivalent to the integrable Hamiltonian system with 2 degrees of freedom in angle-action variables

$$\dot{\sigma} = \partial_I \tilde{\mathcal{H}}, \quad \dot{I} = -\partial_\sigma \tilde{\mathcal{H}}, \quad \dot{\beta} = \partial_K \tilde{\mathcal{H}}, \quad \dot{K} = -\partial_\beta \tilde{\mathcal{H}}$$

with Hamiltonian

$$\tilde{\mathcal{H}}(\sigma, I, \beta, K) = \mathcal{H}(I) + \Omega_2(I)K$$

depending only on the action variables I, K , where $\mathcal{H}(I)$ and $\Omega_2(I)$ are in Theorem 1.1, restricted to the 3-dimensional invariant set $K = 0$.

Remark 1.4 (*Proper invariant subsets*). By Theorem 1.1, the dynamics of system (1.11) on each level set $\mathcal{T}_\ell$ of the pressure is conjugated to a Kronecker flow on $\mathbb{T}^2$ (i.e., a linear flow mod 2π)

$$\dot{\sigma} = \Omega_1, \quad \dot{\beta} = \Omega_2, \tag{1.20}$$

see (1.12). As an immediate consequence, regarding the orbits of the system, we deduce that

- (i) the orbits are closed subsets of $\mathcal{T}_\ell$ if $\Omega_2/\Omega_1 \in \mathbb{Q}$, and dense subsets of $\mathcal{T}_\ell$ if $\Omega_2/\Omega_1 \notin \mathbb{Q}$;
- (ii) each orbit in $\mathcal{T}_\ell$ is a *proper* invariant subset of $\mathcal{T}_\ell$, i.e. it is nonempty and it cannot fill the whole surface $\mathcal{T}_\ell$. The last property is simple to check for (1.20) on $\mathbb{T}^2$: if Ω_1 is nonzero, then the intersection of any orbit with the set $\{0\} \times \mathbb{T}$ of the pairs (σ, β) with $\sigma = 0$ has at most countably many points, and therefore infinitely many points on $\{0\} \times \mathbb{T}$ are outside the orbit; if Ω_1 is zero, then also Ω_2 is zero, and any orbit is a single point;
- (iii) the orbits are *minimal* invariant sets, i.e. any proper subset of an orbit is not an invariant set;
- (iv) any arbitrary union of orbits is also an invariant set.

We conclude that there are no invariant sets other than the orbits themselves (and, trivially, their arbitrary unions).

Remark 1.5 (*Physical interpretation of the action variable*). Item (viii) in Theorem 1.1 gives a physical interpretation of the action variable: up to a constant multiplicative factor, the action I corresponding to a value ℓ of the pressure by (1.18) gives the mass of the fluid that occupies the region E_ℓ enclosed by the pressure level set $\mathcal{T}_\ell$.

Remark 1.6 (*Physical interpretation of the angle variables*). The action variable I has a geometrical, and therefore physical, meaning, see item (viii) in Theorem 1.1 and Remark 1.5. Can we say something similar also for the angle variables σ, β ?

Regarding the angle σ , its physical interpretation is strictly related to the construction of the angle-action coordinates. The full details of the construction are in Sect. 3.7; here we just add some comment.

After passing to cylindrical coordinates $x = \rho \cos \varphi$, $y = \rho \sin \varphi$ in Sect. 3.1, changing the vertical coordinate $z = \tilde{z}/\rho$ (to obtain a canonical Hamiltonian structure) in Sect. 3.2, and introducing symplectic polar coordinates in the radial-vertical plane $\rho = 1 + \sqrt{2\xi} \sin \vartheta$, $\tilde{z} = \sqrt{2\xi} \cos \vartheta$ in Sect. 3.3, we end up with variables $(\vartheta, \varphi, \xi)$ with a rather clear geometrical meaning: a point (x, y, z) in a neighborhood of the circle $\mathcal{C}$ is expressed in terms of $(\vartheta, \varphi, \xi)$ as

$$x = (1 + \sqrt{2\xi} \sin \vartheta) \cos \varphi, \quad y = (1 + \sqrt{2\xi} \sin \vartheta) \sin \varphi, \quad z = \frac{\sqrt{2\xi} \cos \vartheta}{1 + \sqrt{2\xi} \sin \vartheta}.$$

The equations for the variables (ϑ, ξ) form the Hamiltonian system (3.30), which is equivalent to the single equation (3.41) for the angle variable ϑ on the energy level c ; (3.41) is an autonomous scalar equation, and its integration is elementary (Sect. 3.5).

In fact, to construct the angle-action variables, we solve (3.41) with $\chi(c)$ replaced by 1 (the reason is explained in Remark 1.7). After this, and after proving that the solutions $\vartheta(t)$ are periodic functions of time, one realizes that the pair “(time, energy)”, namely s_1^* and c in the lines between (3.49) and (3.51), could be used as new symplectic coordinates: the energy level c determines the orbit, and the time $s_1^* \in [0, T_c^*)$ (where T_c^* is the period, see (3.50)) determines the point on that orbit. Note that here the “energy” is the value of the Hamiltonian $\alpha_3(\vartheta, \xi)$, which is the pressure of the fluid expressed in terms of the coordinates (ϑ, ξ) . However, as is well-known, using the (time, energy) coordinates has two drawbacks:

- (i) even if the time s_1^* can be considered as a circular variable, in fact it is not an angle variable, because it makes a round after a time interval T_c^* , while, for angle variables, a complete round should be 2π ;
- (ii) the frequency $2\pi/T_c^*$ is hidden in the definition of the new coordinates (time, energy), and it does not appear any longer in the transformed system: in fact, the Hamiltonian system in the (time, energy) = (s, c) coordinates becomes $\dot{s} = 1$, $\dot{c} = 0$.

These problems can be easily solved by introducing the normalization factor $2\pi/T_c^*$ multiplying the time variable s_1^* , obtaining a genuine angle variable: this is where the angle variable σ comes from (see the lines between (3.50) and (3.51)). After the circular time variable has been normalized to obtain the geometric angle variable σ , the energy c must also be modified, in order to preserve the canonical Hamiltonian structure of the system: this is the reason for introducing the new variable I , the action, in place of the energy c .

In conclusion, we can say that the physical interpretation of the angle variable σ is this: σ is the product of the time variable of a periodic solution of equation (3.48) (which is the same as equation (3.41) but with $\chi(c)$ replaced by 1) multiplied by the frequency of that solution, so that σ corresponds to a rescaled, normalized version of the time variable of a solution in which the solution takes 2π units to make a complete round and arriving at its starting point.

The other angle, β , is introduced in Sect. 3.8 as a translation of the original horizontal angle φ of the cylindrical coordinates, namely $\varphi = \beta + \eta(\sigma, I)$. From (1.15) and (1.17), the solution (1.14) of the Cauchy problem (1.11), (1.13) expressed in cylindrical coordinates is

$$(x(t), y(t), z(t)) = (\rho(t) \cos \varphi(t), \rho(t) \sin \varphi(t), z(t)),$$

where

$$\begin{aligned} \rho(t) &= \varrho(\sigma(t), I(t)) = \varrho(\sigma_0 + \Omega_1 t, I_0), & z(t) &= \zeta(\sigma(t), I(t)) = \zeta(\sigma_0 + \Omega_1 t, I_0), \\ \varphi(t) &= \beta(t) + \eta(\sigma(t), I(t)) = \beta_0 + \Omega_2 t + \eta(\sigma_0 + \Omega_1 t, I_0), \end{aligned}$$

and, in short, we have written Ω_1, Ω_2 instead of $\Omega_1(I_0), \Omega_2(I_0)$. Thus $\rho(t)$ and $z(t)$ are periodic functions of time with period $2\pi/\Omega_1$, whereas the horizontal angle $\varphi(t)$ is the sum $\beta(t) + \tilde{\eta}(t)$ of the two components

$$\beta(t) = \beta_0 + \Omega_2 t, \quad \tilde{\eta}(t) := \eta(\sigma(t), I(t)) = \eta(\sigma_0 + \Omega_1 t, I_0).$$

If $\Omega_2 \neq 0$, then $\beta(t)$, as a function taking values in $\mathbb{R}$, is a linear function of time, and it is not periodic. For this reason, $\beta(t)$, as a function taking values in $\mathbb{T}$, is a *rotation*. On the contrary, $\tilde{\eta}(t)$ is periodic also as a function taking values in $\mathbb{R}$, because η is 2π -periodic in its first variable, see (3.72). Hence, borrowing a terminology often employed when speaking about the orbits of the pendulum, we could say that $\tilde{\eta}(t)$ is a *libration*.

In conclusion, we can say that the angle β (or, more precisely, the solution $\beta(t)$, which is a function of time) has this physical interpretation: $\beta(t)$ is the rotation component of the motion of the horizontal angle $\varphi(t)$, net of its libration component.

Remark 1.7 (The map Φ is independent of the cut-off function ω). In the definition of the transformations Φ_4 in Sect. 3.7 and Φ_5 in Sect. 3.8, we use a simple, convenient trick: we keep the Gavrilov cut-off function ω (which, after the changes of variables, becomes the factor $\chi(c)$ in (3.41) and $\chi(h(I))$ in (3.70), see (3.7)) out of the construction, using the solutions of the equations one obtains by replacing ω with the constant 1. The gain is that, at the end of the entire construction,

- (i) the map Φ is independent of the cut-off ω and of its support $[\varepsilon, 2\varepsilon]$;
- (ii) the map Φ is defined on the open set $\mathcal{S}^*$, independent on ω, ε , surrounding the circle $\mathcal{C}$, in the sense that $\mathcal{S}^* \cup \mathcal{C}$ is an open neighborhood of $\mathcal{C}$; the set $\mathcal{S}$ (see (1.8)) containing the support of the composition $\omega \circ P$ and of the smooth Gavrillov vector field $\tilde{U}$ (see (1.6)) can be placed in an arbitrary location inside $\mathcal{S}^*$, without affecting the definition and the properties of Φ ;
- (iii) the analytic regularity of the functions $\Psi(\sigma, \beta, \mu)$, $\mathcal{K}(I)$, $\mathcal{R}(I)$, which appear in the formulas of $\Phi(\sigma, \beta, I)$, $\Omega_1(I)$, $\Omega_2(I)$ in Theorem 1.1, has nothing to do with the C^∞ regularity of the cut-off function ω . The analyticity is only limited by the “natural” singularities of the symplectic polar coordinates, i.e., the presence of the square root and the loss of injectivity at the origin. Note that Gavrillov’s functions U, P in (1.4) (the ones before introducing the cut-off ω) are analytic in $\mathcal{N} \setminus \mathcal{C}$, and contain a square root singularity at $\mathcal{C}$: our construction does not deteriorate this regularity, because also the various functions described in Theorem 1.1 are analytic outside the circle $\mathcal{C}$, and have only $\mathcal{C}$ (corresponding to the value $I = 0$ of the action variable) as a possible singular set. $\square$

Remark 1.8 (Relation with Arnold’s structure theorem). By Theorem 1.1, the domain $\mathcal{S}^*$ is foliated by the 2-dimensional tori $\mathcal{T}_\ell$, invariant for the vector field $\tilde{U}$, on which the motion is periodic or quasi-periodic. This is true not only for the Gavrillov vector field, but also for all steady 3D Euler flows whose velocity and vorticity are linearly independent at each point, by an important structure theorem of Arnold, see the papers [1, 2] and the book [3], Chapter II, Section 1.

Thus, Arnold’s result gives the existence of the angle-angle-action variables. On the other hand, since we restrict our study to the specific case of Gavrillov’s vector field, we can go much beyond showing the existence of the variables (σ, β, I) : Theorem 1.1 gives an explicit description of the frequencies $\Omega_1(I)$, $\Omega_2(I)$ close to $I = 0$ (i.e., close to the circle $\mathcal{C}$), proving the existence of both periodic and quasi-periodic motions of the fluid particles.

We remark that, thanks to our detailed knowledge of the Gavrillov vector field (axisymmetric structure and sufficiently high order Taylor expansions), in the proof of Theorem 1.1 we replace Arnold’s general topological argument with the explicit construction of the transformation Φ , leading to a more detailed knowledge about the particle dynamics than in the much more general case covered by Arnold’s structure theorem. We also observe that our proof does not use the vorticity vector field, which, on the contrary, is one of the main ingredients in Arnold’s proof.

Theorem 1.1 is proved in Sects. 3 and 4, splitting the proof into several short simple steps. The proof uses only basic tools from the classical theory of ODEs and dynamical systems, including (Sect. 4.3) the application of the implicit function theorem in a situation of degeneracy—the kind of degeneracy one can expect to find in the construction of the angle-action variables around a strict minimum of the Hamiltonian.

1.3. Related Literature

A general discussion about the existence of compactly supported smooth solutions of PDEs, also in comparison with the result of Gavrillov for the steady Euler equations, can be found in the recent preprint [13]; in particular, for Navier-Stokes equations, see [12].

The nice, explicit construction of Gavrillov’s original paper [10] has been revisited and generalized by Constantin et al. [5]. The paper [5] also uses the result by Gavrillov as a building block to prove the existence of compactly supported solutions of the steady Euler equations that have a given Hölder regularity C^α and are not in C^β for any $\beta > \alpha$. The proof employs the invariances of the Euler equations and the fact that the sum of compactly supported solutions with disjoint supports is itself a solution.

The result by Gavrillov has also been used recently by Enciso, Peralta-Salas and Torres de Lizaur in [8] to produce time-quasi-periodic solutions of the 3D Euler equations. In [8] the authors extend to the 3D case, and to the n -dimensional case for all n even, the construction by Crouseilles and Faou [6] of

time-quasi-periodic solutions of the 2D Euler equations. Both [6, 8] use in a clever way the compactly supported solutions of the steady equations as the main ingredients of the construction.

The time-quasi-periodic solutions in [6, 8] are not of KAM type, that is, they are constructed outside the context of the Kolmogorov–Arnold–Moser perturbation theory of nearly integrable dynamical systems, where small divisor problems typically appear. Time-quasi-periodic solutions of the 3D Euler equations of KAM type are obtained in [4] by Montalto and the author in presence of a forcing term, using pseudo-differential calculus and techniques of KAM theory for PDEs.

As is mentioned in Remark 1.8, the foliation of a region of $\mathbb{R}^3$ by a family of invariant 2-dimensional tori, labelled by the action variable, on which the particle motion is periodic or quasi-periodic, occurs not only for the specific case of the Gavrilov vector field, but also for much more general situations, by Arnold’s structure theorem; see the papers [1, 2] and the book [3], Chapter II, Section 1; see also Remark 1.8 for further comments. Arnold’s result, and its role in the study of the mixing property for Euler flows, is discussed by Khesin et al. [11].

For the existence of periodic motions of fluid particles in steady 3D Euler flows and some related interesting questions, see [9] and Remark 1.13 in Chapter II, Section 1, of [3].

It is reasonable to expect that the proof of Theorem 1.1 can be adapted to obtain an analogous result also for the steady Euler vector fields constructed in [7], which are piecewise smooth, axisymmetric, with compact support in $\mathbb{R}^3$.

2. Dual Role of the Parameter R in Gavrilov’s Solutions

Gavrilov’s construction depends on one parameter, R , but in fact the natural free parameters in his construction are two, not only one. These two natural parameters, called λ and μ in the present section, correspond to two different invariances of the steady Euler equations. The paper [10], without explicitly mentioning it, fixes $\lambda = R$, $\mu = R^2$, namely it considers parameters constrained to stay on the half parabola $(\lambda, \mu) = (\lambda, \lambda^2)$, $\lambda > 0$, instead of pairs of parameters free to vary in the quarter of plane (λ, μ) , $\lambda, \mu > 0$.

In other words, two different roles are played simultaneously by the parameter R in Gavrilov’s solutions, because R is

- both a rescaling factor for the independent variable $(x, y, z) \in \mathbb{R}^3$, appearing as R^{-1} in the argument of α in (1.5),
- and an amplitude factor multiplying the vector fields $U, \tilde{U}$ and the pressures $P, \tilde{P}$, appearing as powers R^3 and R^4 in the definition of b and p in (1.5).

In fact, there exists a family of solutions described by two real parameters (λ, μ) such that the solutions of Sect. 1.1 are obtained in the special case $(\lambda, \mu) = (R, R^2)$. This means that, regarding the parameter R , Gavrilov’s solutions form a 1-parameter subset of a 2-parameter family of solutions. Each element of the family is obtained from any other element of the family by two rescalings, which correspond to two basic invariances of the Euler equations, also preserving the localizability condition. This allows us to study only one element of the family (in particular, a normalized one), obtaining directly a description of all the other elements.

Given $R > 0$, let

$$\mathcal{U} := (\mathcal{C}, \delta, \mathcal{N}, U, P, \varepsilon, \omega, W, \tilde{U}, \tilde{P}, \mathcal{S}, \tau, \mathcal{N}', \mathcal{S}^*) \quad (2.1)$$

denote the list of the elements (sets, constants, and functions) defined in Sect. 1.1. For every $\lambda > 0$ and $\mu > 0$, we define a rescaled version of each element of the list $\mathcal{U}$ in the following way. We define

$$\begin{aligned}
 \mathcal{A}_{\lambda,\mu}\mathcal{C} &:= \{(x, y, z) \in \mathbb{R}^3 : \rho = \lambda R, \ z = 0\}, \\
 \mathcal{A}_{\lambda,\mu}\delta &:= \lambda\delta, \\
 \mathcal{A}_{\lambda,\mu}\mathcal{N} &:= \{(x, y, z) \in \mathbb{R}^3 : (\rho - \lambda R)^2 + z^2 < (\lambda\delta)^2\}, \\
 (\mathcal{A}_{\lambda,\mu}U)(x, y, z) &:= \mu U\left(\frac{x}{\lambda}, \frac{y}{\lambda}, \frac{z}{\lambda}\right) \quad \forall (x, y, z) \in \mathcal{A}_{\lambda,\mu}\mathcal{N}, \\
 (\mathcal{A}_{\lambda,\mu}P)(x, y, z) &:= \mu^2 P\left(\frac{x}{\lambda}, \frac{y}{\lambda}, \frac{z}{\lambda}\right) \quad \forall (x, y, z) \in \mathcal{A}_{\lambda,\mu}\mathcal{N}, \\
 \mathcal{A}_{\lambda,\mu}\varepsilon &:= \mu^2\varepsilon, \\
 (\mathcal{A}_{\lambda,\mu}\omega)(s) &:= \omega\left(\frac{s}{\mu^2}\right) \quad \forall s \in \mathbb{R}, \\
 (\mathcal{A}_{\lambda,\mu}W)(s) &:= \mu^2 W\left(\frac{s}{\mu^2}\right) \quad \forall s \in \mathbb{R}, \\
 (\mathcal{A}_{\lambda,\mu}\tilde{U})(x, y, z) &:= \mu\tilde{U}\left(\frac{x}{\lambda}, \frac{y}{\lambda}, \frac{z}{\lambda}\right) \quad \forall (x, y, z) \in \mathbb{R}^3, \\
 (\mathcal{A}_{\lambda,\mu}\tilde{P})(x, y, z) &:= \mu^2\tilde{P}\left(\frac{x}{\lambda}, \frac{y}{\lambda}, \frac{z}{\lambda}\right), \quad \forall (x, y, z) \in \mathbb{R}^3, \\
 \mathcal{A}_{\lambda,\mu}\mathcal{S} &:= \{(x, y, z) \in \mathcal{A}_{\lambda,\mu}\mathcal{N} : \mathcal{A}_{\lambda,\mu}\varepsilon < (\mathcal{A}_{\lambda,\mu}P)(x, y, z) < 2\mathcal{A}_{\lambda,\mu}\varepsilon\}, \\
 \mathcal{A}_{\lambda,\mu}\tau &:= \mu^2\tau, \\
 \mathcal{A}_{\lambda,\mu}\mathcal{N}' &:= \{(x, y, z) \in \mathbb{R}^3 : (\rho - \lambda R)^2 + z^2 < (\lambda\delta/4)^2\}, \\
 \mathcal{A}_{\lambda,\mu}\mathcal{S}^* &:= \{(x, y, z) \in \mathcal{A}_{\lambda,\mu}\mathcal{N} : 0 < (\mathcal{A}_{\lambda,\mu}P)(x, y, z) < 3\mathcal{A}_{\lambda,\mu}\varepsilon\},
 \end{aligned} \tag{2.2}$$

where $\rho = \sqrt{x^2 + y^2}$. We denote by $\mathcal{A}_{\lambda,\mu}\mathcal{U} := (\mathcal{A}_{\lambda,\mu}\mathcal{C}, \dots, \mathcal{A}_{\lambda,\mu}\mathcal{S}^*)$ the list of the rescaled elements. The properties of $\mathcal{U}$ in Sect. 1.1 become the following properties for $\mathcal{A}_{\lambda,\mu}\mathcal{U}$.

- The constant $\mathcal{A}_{\lambda,\mu}\delta = \lambda\delta$ satisfies $0 < \lambda\delta < \lambda R$ and $\lambda\delta \leq \lambda r_0 R$.
- The rescaled pressure $\mathcal{A}_{\lambda,\mu}P$ is analytic in $\mathcal{A}_{\lambda,\mu}\mathcal{N}$.
- The rescaled vector field $\mathcal{A}_{\lambda,\mu}U$ is continuous in $\mathcal{A}_{\lambda,\mu}\mathcal{N}$ and analytic in $(\mathcal{A}_{\lambda,\mu}\mathcal{N}) \setminus (\mathcal{A}_{\lambda,\mu}\mathcal{C})$.
- The pair $(\mathcal{A}_{\lambda,\mu}U, \mathcal{A}_{\lambda,\mu}P)$ satisfies the Euler equations and the localizability condition in $(\mathcal{A}_{\lambda,\mu}\mathcal{N}) \setminus (\mathcal{A}_{\lambda,\mu}\mathcal{C})$.
- The function $\mathcal{A}_{\lambda,\mu}\omega$ is $C^\infty(\mathbb{R}, \mathbb{R})$ with support contained in $[\mathcal{A}_{\lambda,\mu}\varepsilon, 2\mathcal{A}_{\lambda,\mu}\varepsilon] = [\mu^2\varepsilon, 2\mu^2\varepsilon]$.
- The function $\mathcal{A}_{\lambda,\mu}W$ satisfies

$$(\mathcal{A}_{\lambda,\mu}W)(s) = \int_0^s (\mathcal{A}_{\lambda,\mu}\omega)^2(\sigma) d\sigma \quad \forall s \in \mathbb{R}.$$

- The vector field $\mathcal{A}_{\lambda,\mu}\tilde{U}$ satisfies

$$\begin{aligned}
 (\mathcal{A}_{\lambda,\mu}\tilde{U})(x, y, z) &= \mu\omega\left(P\left(\frac{x}{\lambda}, \frac{y}{\lambda}, \frac{z}{\lambda}\right)\right)U\left(\frac{x}{\lambda}, \frac{y}{\lambda}, \frac{z}{\lambda}\right) \\
 &= (\mathcal{A}_{\lambda,\mu}\omega)((\mathcal{A}_{\lambda,\mu}P)(x, y, z)) \cdot (\mathcal{A}_{\lambda,\mu}U)(x, y, z) \quad \forall (x, y, z) \in \mathcal{A}_{\lambda,\mu}\mathcal{N}
 \end{aligned}$$

and $(\mathcal{A}_{\lambda,\mu}\tilde{U})(x, y, z) = 0$ for all $(x, y, z) \in \mathbb{R}^3 \setminus (\mathcal{A}_{\lambda,\mu}\mathcal{N})$.

- The pressure $\mathcal{A}_{\lambda,\mu}\tilde{P}$ satisfies

$$(\mathcal{A}_{\lambda,\mu}\tilde{P})(x, y, z) = \mu^2 W\left(P\left(\frac{x}{\lambda}, \frac{y}{\lambda}, \frac{z}{\lambda}\right)\right) = (\mathcal{A}_{\lambda,\mu}W)((\mathcal{A}_{\lambda,\mu}P)(x, y, z)) \quad \forall (x, y, z) \in \mathcal{A}_{\lambda,\mu}\mathcal{N}$$

and $(\mathcal{A}_{\lambda,\mu}\tilde{P})(x, y, z) = \mu^2 W(2\varepsilon) = (\mathcal{A}_{\lambda,\mu}W)(2\mathcal{A}_{\lambda,\mu}\varepsilon)$ for all $(x, y, z) \in \mathbb{R}^3 \setminus (\mathcal{A}_{\lambda,\mu}\mathcal{N})$.

- Both $\mathcal{A}_{\lambda,\mu}\tilde{U}$ and $\mathcal{A}_{\lambda,\mu}\tilde{P}$ are $C^\infty(\mathbb{R}^3)$ and satisfy the Euler equations and the localizability condition in $\mathbb{R}^3$.
- Both $\mathcal{A}_{\lambda,\mu}\tilde{U}$ and $\nabla(\mathcal{A}_{\lambda,\mu}\tilde{P})$ vanish outside the bounded set $\mathcal{A}_{\lambda,\mu}\mathcal{S}$.

- One has $\mathcal{A}_{\lambda,\mu}P > \mathcal{A}_{\lambda,\mu}\tau$ in $(\mathcal{A}_{\lambda,\mu}\mathcal{N}) \setminus (\mathcal{A}_{\lambda,\mu}\mathcal{N}')$.
- One has $\mathcal{A}_{\lambda,\mu}\tau \geq 3\mathcal{A}_{\lambda,\mu}\varepsilon$.

Thus, we have obtained the 2-parameter family $\{\mathcal{A}_{\lambda,\mu}\mathcal{U}\}_{\lambda,\mu}$. One has the group property

$$\mathcal{A}_{\lambda_1,\mu_1}(\mathcal{A}_{\lambda_2,\mu_2}\mathcal{U}) = \mathcal{A}_{\lambda_1\lambda_2,\mu_1\mu_2}\mathcal{U}, \quad \mathcal{A}_{\lambda,\mu}(\mathcal{A}_{\frac{1}{\lambda},\frac{1}{\mu}}\mathcal{U}) = \mathcal{A}_{1,1}\mathcal{U} = \mathcal{U} \quad (2.3)$$

for all $\lambda_1, \lambda_2, \lambda, \mu_1, \mu_2, \mu \in (0, \infty)$. The check of (2.3) is straightforward.

Lemma 2.1. *Given $R > 0$, let $\mathcal{U}_R$ be the list (2.1) of the elements (sets, constants, and functions) defined in Sect. 1.1. Then*

$$\mathcal{U}_R = \mathcal{A}_{R,R^2}\mathcal{U}_1$$

where $\mathcal{U}_1$ is a list of elements constructed in Sect. 1.1 for $R = 1$, and $\mathcal{A}_{R,R^2}$ is the rescaling operator $\mathcal{A}_{\lambda,\mu}$, defined in (2.2), at $\lambda = R$, $\mu = R^2$.

Proof. Let $\mathcal{U}_R$ be the list (2.1) of elements constructed in Sect. 1.1 for a given $R > 0$. We observe that the list $\mathcal{A}_{\lambda,\mu}\mathcal{U}_R$ with $\lambda = 1/R$ and $\mu = 1/R^2$ coincides with a list of elements that one obtains by choosing $R = 1$ in Sect. 1.1, which we call $\mathcal{U}_1$. The check is elementary; for example, regarding the vector field U and the pressure P , by (1.4) and (1.5) one has

$$\begin{aligned} \frac{u_\rho(R\rho, Rz)}{R^2} &= \frac{\partial_z \alpha(\rho, z)}{4\rho}, & \frac{u_\varphi(R\rho, Rz)}{R^2} &= \frac{\sqrt{H(\alpha(\rho, z))}}{4\rho}, \\ \frac{u_z(R\rho, Rz)}{R^2} &= -\frac{\partial_\rho \alpha(\rho, z)}{4\rho}, & \frac{P(Rx, Ry, Rz)}{R^4} &= \frac{\alpha(\rho, z)}{4}. \end{aligned}$$

Then, by (2.3), $\mathcal{U}_R = \mathcal{A}_{R,R^2}(\mathcal{A}_{\frac{1}{R},\frac{1}{R^2}}\mathcal{U}_R) = \mathcal{A}_{R,R^2}\mathcal{U}_1$. $\square$

Regarding the fluid particle system (1.11), the consequence of Lemma 2.1 is the following lemma, whose proof is trivial.

Lemma 2.2. *Let $\tilde{\mathcal{U}}_R = \mathcal{A}_{R,R^2}\tilde{\mathcal{U}}_1$, where $\tilde{\mathcal{U}}_R$ is given by Sect. 1.1 for some $R > 0$ and $\tilde{\mathcal{U}}_1$ is given by Sect. 1.1 for $R = 1$. Then a function $(x(t), y(t), z(t))$ solves the fluid particle system (1.11) with velocity field $\tilde{U} = \tilde{\mathcal{U}}_R$ if and only if*

$$x(t) = Rx_1(Rt), \quad y(t) = Ry_1(Rt), \quad z(t) = Rz_1(Rt),$$

where $(x_1(t), y_1(t), z_1(t))$ solves

$$(\dot{x}_1(t), \dot{y}_1(t), \dot{z}_1(t)) = \tilde{\mathcal{U}}_1(x_1(t), y_1(t), z_1(t)).$$

3. Conjugation to a Linear Flow on $\mathbb{T}^2$

In this section and in Sect. 4 we prove Theorem 1.1. Thus, assume that $\mathcal{U}$ in (2.1) is given by Sect. 1.1 for $R = 1$. Hence, in particular,

$$\begin{aligned} \mathcal{C} &= \{(x, y, z) \in \mathbb{R}^3 : \rho = 1, z = 0\}, \\ \mathcal{N} &= \{(x, y, z) \in \mathbb{R}^3 : (\rho - 1)^2 + z^2 < \delta^2\}, \\ \mathcal{N}' &= \{(x, y, z) \in \mathbb{R}^3 : (\rho - 1)^2 + z^2 < (\delta/4)^2\}, \end{aligned} \quad (3.1)$$

where $\rho = \sqrt{x^2 + y^2}$. The constant δ satisfies $0 < \delta < 1$ and $\delta \leq r_0$. The vector field U and the pressure P in $\mathcal{N}$ are given by (1.4), (1.5), where $a(\rho, z) = \alpha(\rho, z)$ and

$$p(\rho, z) = \frac{\alpha(\rho, z)}{4}, \quad b(\rho, z) = \frac{1}{4}\sqrt{H(\alpha(\rho, z))}. \quad (3.2)$$

The constants τ and ε satisfy (1.9) and $\tau \geq 3\varepsilon$. The function ω is supported in $[\varepsilon, 2\varepsilon]$, and W is in (1.7). The vector field $\tilde{U}$ and the pressure $\tilde{P}$ are given by (1.6) in $\mathcal{N}$, and $(\tilde{U}, \tilde{P}) = (0, W(2\varepsilon))$ in $\mathbb{R}^3 \setminus \mathcal{N}$. The sets $\mathcal{S}, \mathcal{S}^*$ are given by (1.8), (1.10).

Outside the set $\mathcal{S}$, the vector field $\tilde{U}$ is zero, and the solutions of system (1.11) are constant in time. Hence, by the uniqueness property of the solution of Cauchy problems, any solution of (1.11) that is in $\mathcal{S}$ at some time t_0 remains in $\mathcal{S}$ for its entire lifespan; in other words, $\mathcal{S}$ is an invariant set for (1.11). Moreover $\mathcal{S}$ is bounded, and therefore, by basic ODE theory, the solutions of (1.11) in $\mathcal{S}$ are all global in time.

Trivially, any subset of $\mathbb{R}^3 \setminus \mathcal{S}$ is also an invariant set for (1.11). Hence any subset of $\mathbb{R}^3$ containing $\mathcal{S}$ is invariant for (1.11). In particular, $\mathcal{S}^*$ and $\mathcal{N}$ are invariant for (1.11).

3.1. Cylindrical Coordinates

To study system (1.11) in $\mathcal{N}$, first of all we move on to cylindrical coordinates, that is, we consider the diffeomorphism

$$\begin{aligned}\Phi_1 : \mathcal{N}_1 &\rightarrow \mathcal{N}, & \Phi_1(\rho, \varphi, z) &:= (\rho \cos \varphi, \rho \sin \varphi, z), \\ \mathcal{N}_1 &:= \{(\rho, \varphi, z) \in \mathbb{R} \times \mathbb{T} \times \mathbb{R} : (\rho - 1)^2 + z^2 < \delta^2\},\end{aligned}\quad (3.3)$$

and the change of variables $\tilde{v} = \Phi_1(v)$, with $\tilde{v} = (x, y, z)$, $v = (\rho, \varphi, z)$, i.e., $x = \rho \cos \varphi$, $y = \rho \sin \varphi$. Using cylindrical coordinates is a natural choice because the quantities $u_\rho, u_\varphi, u_z, p$ in (1.5) are already expressed in terms of ρ, z . Now a function $\tilde{v}(t) = \Phi_1(v(t))$ solves (1.11) in $\mathcal{N}$ if and only if $v(t)$ solves

$$\dot{v}(t) = V_1(v(t)) \quad (3.4)$$

in $\mathcal{N}_1$, where V_1 is the vector field

$$V_1(v) := (D\Phi_1(v))^{-1} \tilde{U}(\Phi_1(v)), \quad v = (\rho, \varphi, z) \in \mathcal{N}_1.$$

The Jacobian matrix $D\Phi_1(v)$ and its inverse matrix are

$$D\Phi_1(v) = \begin{bmatrix} \cos \varphi & -\rho \sin \varphi & 0 \\ \sin \varphi & \rho \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (D\Phi_1(v))^{-1} = \begin{bmatrix} \cos \varphi & \sin \varphi & 0 \\ -\frac{1}{\rho} \sin \varphi & \frac{1}{\rho} \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (3.5)$$

the composition $\tilde{U}(\Phi_1(v))$ is

$$\tilde{U}(\Phi_1(v)) = \omega(p(\rho, z))U(\Phi_1(v)), \quad U(\Phi_1(v)) = \begin{pmatrix} u_\rho(\rho, z) \cos \varphi - u_\varphi(\rho, z) \sin \varphi \\ u_\rho(\rho, z) \sin \varphi + u_\varphi(\rho, z) \cos \varphi \\ u_z(\rho, z) \end{pmatrix},$$

and therefore

$$\begin{aligned}V_1(\rho, \varphi, z) &= \omega(p(\rho, z)) \left(u_\rho(\rho, z), \frac{u_\varphi(\rho, z)}{\rho}, u_z(\rho, z) \right) \\ &= \left(\frac{\omega(p(\rho, z)) \partial_z p(\rho, z)}{\rho}, \frac{\omega(p(\rho, z)) b(\rho, z)}{\rho^2}, -\frac{\omega(p(\rho, z)) \partial_\rho p(\rho, z)}{\rho} \right).\end{aligned}\quad (3.6)$$

Since α and H are the functions constructed and studied in [10], it is convenient to express the other quantities in terms of them. Hence, we define

$$\chi(s) := \frac{1}{4} \omega\left(\frac{s}{4}\right) \quad \forall s \in \mathbb{R}, \quad (3.7)$$

and, recalling (3.2), we rewrite (3.6) as

$$V_1(\rho, \varphi, z) = \chi(\alpha(\rho, z)) \left(\frac{\partial_z \alpha(\rho, z)}{\rho}, \frac{\sqrt{H(\alpha(\rho, z))}}{\rho^2}, -\frac{\partial_\rho \alpha(\rho, z)}{\rho} \right). \quad (3.8)$$

The curve $\mathcal{C}$ and the sets $\mathcal{S}^*, \mathcal{N}'$ become

$$\begin{aligned}\mathcal{C}_1 &:= \Phi_1^{-1}(\mathcal{C}) = \{(1, \varphi, 0) : \varphi \in \mathbb{T}\}, \\ \mathcal{S}_1^* &:= \Phi_1^{-1}(\mathcal{S}^*) = \{(\rho, \varphi, z) \in \mathcal{N}_1 : 0 < \alpha(\rho, z) < 4\tau\}, \\ \mathcal{N}'_1 &:= \Phi_1^{-1}(\mathcal{N}') = \{(\rho, \varphi, z) \in \mathcal{N}_1 : (\rho - 1)^2 + z^2 < (\delta/4)^2\}.\end{aligned}$$

By (1.9) and (3.2), one has

$$\alpha(\rho, z) > 4\tau \quad \forall (\rho, z) \in \mathcal{N}_1 \setminus \mathcal{N}'_1. \quad (3.9)$$

Moreover, the map Φ_1 is analytic in $\mathcal{N}_1$.

The vector field U satisfies the localizability condition (1.2) in $\mathcal{N} \setminus \mathcal{C}$, and therefore $\tilde{U} \cdot \nabla P = 0$ in $\mathcal{N}$. Hence the pressure P is a prime integral of system (1.11) in $\mathcal{N}$. In cylindrical coordinates, this means that $p(\rho, z)$, and therefore $\alpha(\rho, z)$ too, are prime integrals of (3.4) in $\mathcal{N}_1$. This can also be verified directly: by (3.4) and (3.8),

$$\frac{d}{dt} \{\alpha(\rho(t), z(t))\} = \partial_\rho \alpha(\rho, z) \dot{\rho} + \partial_z \alpha(\rho, z) \dot{z} = 0.$$

Hence every trajectory $\{v(t) = (\rho(t), \varphi(t), z(t)) : t \in \mathbb{R}\}$ of system (3.4) in $\mathcal{N}_1$ lies in a level set

$$\mathcal{P}_c := \{(\rho, \varphi, z) \in \mathcal{N}_1 : \alpha(\rho, z) = c\}. \quad (3.10)$$

In particular, every trajectory of (3.4) in $\mathcal{S}_1^*$ lies in a level set $\mathcal{P}_c$ with $0 < c < 4\tau$. In fact, the set $\mathcal{S}_1^*$ is exactly the union of all the level sets $\mathcal{P}_c$ with $c \in (0, 4\tau)$.

3.2. Elimination of the Factor $1/\rho$ and Canonical Hamiltonian Structure

We want to remove the factor $1/\rho$ appearing the first and third component of V_1 in (3.6). We consider the diffeomorphism

$$\begin{aligned}\Phi_2 : \mathcal{N}_2 &\rightarrow \mathcal{N}_1, \quad \Phi_2(\rho, \varphi, z) = \left(\rho, \varphi, \frac{z}{\rho}\right), \\ \mathcal{N}_2 &:= \{(\rho, \varphi, z) \in \mathbb{R} \times \mathbb{T} \times \mathbb{R} : (\rho - 1)^2 + z^2 \rho^{-2} < \delta^2\},\end{aligned} \quad (3.11)$$

and the change of variables $\tilde{v} = \Phi_2(v)$, with $\tilde{v} = (\tilde{\rho}, \tilde{\varphi}, \tilde{z})$, $v = (\rho, \varphi, z)$. A function $\tilde{v}(t) = \Phi_2(v(t))$ solves (3.4) in $\mathcal{N}_1$ if and only if $v(t)$ solves

$$\dot{v}(t) = V_2(v(t)) \quad (3.12)$$

in $\mathcal{N}_2$, where

$$V_2(v) := (D\Phi_2(v))^{-1} V_1(\Phi_2(v)), \quad v = (\rho, \varphi, z) \in \mathcal{N}_2. \quad (3.13)$$

The Jacobian matrix $D\Phi_2(v)$ and its inverse are

$$D\Phi_2(v) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -z\rho^{-2} & 0 & \rho^{-1} \end{bmatrix}, \quad (D\Phi_2(v))^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ z\rho^{-1} & 0 & \rho \end{bmatrix}. \quad (3.14)$$

We define

$$\alpha_2(\rho, z) := \alpha\left(\rho, \frac{z}{\rho}\right) \quad (3.15)$$

for all $(\rho, z) \in \mathcal{D}_2$. Hence

$$\partial_z \alpha_2(\rho, z) = (\partial_z \alpha)\left(\rho, \frac{z}{\rho}\right) \frac{1}{\rho}, \quad \partial_\rho \alpha_2(\rho, z) = (\partial_\rho \alpha)\left(\rho, \frac{z}{\rho}\right) - (\partial_z \alpha)\left(\rho, \frac{z}{\rho}\right) \frac{z}{\rho^2},$$

and, recalling the second identity in (3.2),

$$V_1(\Phi_2(\rho, \varphi, z)) = \chi(\alpha_2(\rho, z)) \left(\partial_z \alpha_2(\rho, z), \frac{\sqrt{H(\alpha_2(\rho, z))}}{\rho^2}, -\frac{\partial_\rho \alpha_2(\rho, z)}{\rho} - \frac{z \partial_z \alpha_2(\rho, z)}{\rho^2} \right).$$

Then the vector field V_2 in (3.13) is

$$V_2(\rho, \varphi, z) = \chi(\alpha_2(\rho, z)) \left(\partial_z \alpha_2(\rho, z), \frac{\sqrt{H(\alpha_2(\rho, z))}}{\rho^2}, -\partial_\rho \alpha_2(\rho, z) \right).$$

The sets $\mathcal{C}_1, \mathcal{S}_1^*, \mathcal{N}'_1$ become

$$\begin{aligned} \mathcal{C}_2 &:= \Phi_2^{-1}(\mathcal{C}_1) = \{(1, \varphi, 0) : \varphi \in \mathbb{T}\} = \mathcal{C}_1, \\ \mathcal{S}_2^* &:= \Phi_2^{-1}(\mathcal{S}_1^*) = \{(\rho, \varphi, z) \in \mathcal{N}_2 : 0 < \alpha_2(\rho, z) < 4\tau\}, \\ \mathcal{N}'_2 &:= \Phi_2^{-1}(\mathcal{N}'_1) = \{(\rho, \varphi, z) \in \mathcal{N}_2 : (\rho - 1)^2 + z^2 \rho^{-2} < (\delta/4)^2\}. \end{aligned}$$

Note that Φ_2 leaves $\mathcal{C}_2 = \mathcal{C}_1$ invariant, because $z/\rho = z$ at $\rho = 1$. By (3.9), one has

$$\alpha_2(\rho, z) > 4\tau \quad \forall (\rho, \varphi, z) \in \mathcal{N}_2 \setminus \mathcal{N}'_2. \quad (3.16)$$

The level set $\mathcal{P}_c$ of α in (3.10) becomes the level set

$$\mathcal{P}_{2,c} = \{(\rho, \varphi, z) \in \mathcal{N}_2 : \alpha_2(\rho, z) = c\} \quad (3.17)$$

of α_2 . The set $\mathcal{S}_2^*$ is the union of the level sets $\mathcal{P}_{2,c}$ with $c \in (0, 4\tau)$. The map Φ_2 is analytic in $\mathcal{N}_2$. The function $\alpha_2(\rho, z)$ is a prime integral of system (3.12), and it is also analytic; its Taylor expansion around $(1, 0)$ is in (4.14). The first and third equation of system (3.12) are the Hamiltonian system

$$\dot{\rho} = \partial_z \mathcal{H}_2(\rho, z), \quad \dot{z} = -\partial_\rho \mathcal{H}_2(\rho, z), \quad (3.18)$$

where

$$\mathcal{H}_2(\rho, z) := \Gamma(\alpha_2(\rho, z)), \quad \Gamma(s) := \int_0^s \chi(\sigma) d\sigma. \quad (3.19)$$

3.3. Symplectic Polar Coordinates in the Radial-Vertical Plane

Now we move on to polar coordinates (in their symplectic version) to describe the pair (ρ, z) . The pairs (ρ, z) such that $(\rho, \varphi, z) \in \mathcal{N}_2$ do not form a disc; thus, it is convenient to consider a subset of $\mathcal{N}_2$ that fits with polar coordinates better than how $\mathcal{N}_2$ does. We consider the open sets

$$\begin{aligned} \mathcal{B}_2 &:= \{(\rho, \varphi, z) \in \mathbb{R} \times \mathbb{T} \times \mathbb{R} : 0 < (\rho - 1)^2 + z^2 < \delta_2^2\}, \\ \mathcal{B}'_2 &:= \{(\rho, \varphi, z) \in \mathbb{R} \times \mathbb{T} \times \mathbb{R} : 0 < (\rho - 1)^2 + z^2 < (\delta_2/2)^2\}, \end{aligned} \quad (3.20)$$

where $\delta_2 = C\delta$, $C > 0$, and we observe that

$$\mathcal{B}_2 \subseteq (\mathcal{N}_2 \setminus \mathcal{C}_2) \subseteq \mathcal{N}_2, \quad (\mathcal{N}'_2 \setminus \mathcal{C}_2) \subseteq \mathcal{B}'_2, \quad (\mathcal{B}_2 \setminus \mathcal{B}'_2) \subseteq (\mathcal{N}_2 \setminus \mathcal{N}'_2) \quad (3.21)$$

if $C(1 + \delta) \leq 1$ and $1 + (\delta/4) \leq 2C$. This holds, for example, for $C = 2/3$ and all $0 < \delta \leq 1/2$.

Proof. (Proof of (3.21)) Let $\delta_2 = C\delta$, let $C(1 + \delta) \leq 1$, and $(\rho, \varphi, z) \in \mathcal{B}_2$. Then $(\rho - 1)^2 < \delta_2^2$, whence $\rho > 1 - \delta_2$. Moreover $1 - \delta_2 = 1 - C\delta \geq C > 0$. Hence

$$(\rho - 1)^2 + \frac{z^2}{\rho^2} \leq (\rho - 1)^2 + \frac{z^2}{(1 - \delta_2)^2} \leq \frac{(\rho - 1)^2 + z^2}{(1 - \delta_2)^2} < \frac{\delta_2^2}{(1 - \delta_2)^2} \leq \delta^2,$$

where the last inequality holds because $C(1 + \delta) \leq 1$. Therefore $(\rho, \varphi, z) \in \mathcal{N}_2$. Moreover $0 < (\rho - 1)^2 + z^2$. This proves that $\mathcal{B}_2 \subseteq (\mathcal{N}_2 \setminus \mathcal{C}_2)$.

Now let $\delta_2 = C\delta$, let $1 + (\delta/4) \leq 2C$, and $(\rho, \varphi, z) \in \mathcal{N}'_2 \setminus \mathcal{C}_2$. Since $(\rho, \varphi, z) \in \mathcal{N}'_2$, one has $(\rho - 1)^2 < (\delta/4)^2$, whence $\rho < 1 + (\delta/4)$, and

$$(\rho - 1)^2 + z^2 \leq \left((\rho - 1)^2 + \frac{z^2}{\rho^2} \right) \left(1 + \frac{\delta}{4} \right)^2 < \frac{\delta^2}{16} \left(1 + \frac{\delta}{4} \right)^2 \leq \frac{\delta^2}{4},$$

where the last inequality holds because $1 + (\delta/4) \leq 2C$. Also, $0 < (\rho - 1)^2 + z^2$ because $(\rho, \varphi, z) \notin \mathcal{C}_2$. Therefore $(\rho, \varphi, z) \in \mathcal{B}'_2$. This proves that $(\mathcal{N}'_2 \setminus \mathcal{C}_2) \subseteq \mathcal{B}'_2$.

The last inclusion in (3.21) holds because $\mathcal{C}_2 \subseteq \mathcal{N}'_2 \subseteq \mathcal{N}_2$ and $(\mathcal{N}_2 \setminus \mathcal{C}_2) \setminus (\mathcal{N}'_2 \setminus \mathcal{C}_2) = (\mathcal{N}_2 \setminus \mathcal{N}'_2)$. $\square$

By (3.21) and (3.16), one has

$$\alpha_2(\rho, z) > 4\tau \quad \forall (\rho, \varphi, z) \in \mathcal{B}_2 \setminus \mathcal{B}'_2. \quad (3.22)$$

Thus $\mathcal{S}_2^* \subseteq \mathcal{B}'_2 \subseteq \mathcal{B}_2$, and they are invariant sets for system (3.12).

We consider the diffeomorphism

$$\Phi_3 : \mathcal{B}_3 \rightarrow \mathcal{B}_2, \quad \Phi_3(\vartheta, \varphi, \xi) = (1 + \sqrt{2\xi} \sin \vartheta, \varphi, \sqrt{2\xi} \cos \vartheta), \quad \mathcal{B}_3 := \mathbb{T} \times \mathbb{T} \times (0, \xi_3), \quad (3.23)$$

where $\xi_3 := \delta_2^2/2 = 2\delta^2/9$, and the change of variables $\tilde{v} = \Phi_3(v)$, with $\tilde{v} = (\tilde{\rho}, \tilde{\varphi}, \tilde{z})$, $v = (\vartheta, \varphi, \xi)$. A function $\tilde{v}(t) = \Phi_3(v(t))$ solves (3.12) in $\mathcal{B}_2$ if and only if $v(t)$ solves

$$\dot{v}(t) = V_3(v(t)) \quad (3.24)$$

in $\mathcal{B}_3$, where

$$V_3(v) := (D\Phi_3(v))^{-1}V_2(\Phi_3(v)), \quad v = (\vartheta, \varphi, \xi) \in \mathcal{B}_3.$$

The Jacobian matrix $D\Phi_3(v)$ and its inverse are

$$D\Phi_3(v) = \begin{bmatrix} \sqrt{2\xi} \cos \vartheta & 0 & \frac{1}{\sqrt{2\xi}} \sin \vartheta \\ 0 & 1 & 0 \\ -\sqrt{2\xi} \sin \vartheta & 0 & \frac{1}{\sqrt{2\xi}} \cos \vartheta \end{bmatrix}, \quad (D\Phi_3(v))^{-1} = \begin{bmatrix} \frac{1}{\sqrt{2\xi}} \cos \vartheta & 0 & -\frac{1}{\sqrt{2\xi}} \sin \vartheta \\ 0 & 1 & 0 \\ \sqrt{2\xi} \sin \vartheta & 0 & \sqrt{2\xi} \cos \vartheta \end{bmatrix}. \quad (3.25)$$

We define

$$\alpha_3(\vartheta, \xi) := \alpha_2(1 + \sqrt{2\xi} \sin \vartheta, \sqrt{2\xi} \cos \vartheta), \quad (3.26)$$

for all $(\vartheta, \xi) \in \mathbb{T} \times [0, \xi_3]$. In fact, in the set $\mathcal{B}_3$, ξ varies in the interval $(0, \xi_3)$, but it is convenient to consider α_3 also for $\xi = 0$; one has $\alpha_3(\vartheta, 0) = \alpha_2(1, 0) = \alpha(1, 0) = 0$ (see (3.15)). From (3.26) it follows that

$$\begin{aligned} \partial_{\vartheta} \alpha_3(\vartheta, \xi) &= (\partial_{\rho} \alpha_2)(\rho, z) \sqrt{2\xi} \cos \vartheta - (\partial_z \alpha_2)(\rho, z) \sqrt{2\xi} \sin \vartheta, \\ \partial_{\xi} \alpha_3(\vartheta, \xi) &= (\partial_{\rho} \alpha_2)(\rho, z) \frac{1}{\sqrt{2\xi}} \sin \vartheta + (\partial_z \alpha_2)(\rho, z) \frac{1}{\sqrt{2\xi}} \cos \vartheta \end{aligned} \quad (3.27)$$

for all $(\vartheta, \xi) \in \mathbb{T} \times (0, \xi_3)$, where $(\rho, z) = (1 + \sqrt{2\xi} \sin \vartheta, \sqrt{2\xi} \cos \vartheta)$. Hence

$$V_3(\vartheta, \varphi, \xi) = \chi(\alpha_3(\vartheta, \xi)) \left(\partial_{\xi} \alpha_3(\vartheta, \xi), \frac{\sqrt{H(\alpha_3(\vartheta, \xi))}}{(1 + \sqrt{2\xi} \sin \vartheta)^2}, -\partial_{\vartheta} \alpha_3(\vartheta, \xi) \right). \quad (3.28)$$

The set $\mathcal{C}_2$ is out of $\mathcal{B}_2$; the sets $\mathcal{S}_2^*$, $\mathcal{B}'_2$ become

$$\begin{aligned} \mathcal{S}_3^* &:= \Phi_3^{-1}(\mathcal{S}_2^*) = \{(\vartheta, \varphi, \xi) \in \mathcal{B}_3 : 0 < \alpha_3(\vartheta, \xi) < 4\tau\}, \\ \mathcal{B}'_3 &:= \Phi_3^{-1}(\mathcal{B}'_2) = \mathbb{T} \times \mathbb{T} \times (0, \xi_3/4). \end{aligned}$$

By (3.22), one has

$$\alpha_3(\vartheta, \xi) > 4\tau \quad \forall (\vartheta, \varphi, \xi) \in \mathcal{B}_3 \setminus \mathcal{B}'_3 = \mathbb{T} \times \mathbb{T} \times [\xi_3/4, \xi_3]. \quad (3.29)$$

Thus $\mathcal{S}_3^* \subset \mathcal{B}'_3$. The level set $\mathcal{P}_{2,c}$ of α_2 in (3.17) becomes the level set

$$\mathcal{P}_{3,c} = \{(\vartheta, \varphi, \xi) \in \mathcal{B}_3 : \alpha_3(\vartheta, \xi) = c\}$$

of α_3 . The set $\mathcal{S}_3^*$ is the union of the level sets $\mathcal{P}_{3,c}$ with $c \in (0, 4\tau)$. The map Φ_3 is analytic in $\mathcal{B}_3$. The function $\alpha_3(\vartheta, \xi)$ is a prime integral of system (3.24); its behavior near $\xi = 0$ is studied in Sects. 3.4 and 4.3. The first and third equation of system (3.24) are the Hamiltonian system

$$\dot{\vartheta} = \partial_{\xi} \mathcal{H}_3(\vartheta, \xi), \quad \dot{\xi} = -\partial_{\vartheta} \mathcal{H}_3(\vartheta, \xi), \quad (3.30)$$

where

$$\mathcal{H}_3(\vartheta, \xi) := \mathcal{H}_2(1 + \sqrt{2\xi} \sin \vartheta, \sqrt{2\xi} \cos \vartheta) = \Gamma(\alpha_3(\vartheta, \xi)), \quad (3.31)$$

where $\mathcal{H}_2$ and Γ are defined in (3.19). Moreover ϑ (as well as φ) is an angle variable, i.e., it varies in $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$.

- Remark 3.1.* (i) We use the symplectic transformation $(\rho, z) = (1 + \sqrt{2\xi} \sin \vartheta, \sqrt{2\xi} \cos \vartheta)$, instead of the simpler polar coordinates $(\rho, z) = (1 + r \cos \vartheta, r \sin \vartheta)$ or $(\rho, z) = (1 + r \sin \vartheta, r \cos \vartheta)$, in order to preserve the canonical Hamiltonian structure of system (3.18).
- (ii) Using $\cos \vartheta$ for z and $\sin \vartheta$ for ρ in the definition (3.23) of Φ_3 , instead of vice versa, is just a matter of convenience: in this way, we get a positive angular velocity for the solution $\vartheta(t)$.

3.4. Level Sets

Since the square root function $\xi \mapsto \sqrt{\xi}$ is analytic in $(0, \infty)$, the function $\alpha_3(\vartheta, \xi)$ defined in (3.26) is analytic in $(\vartheta, \xi) \in \mathbb{T} \times (0, \xi_3)$. By the Taylor expansion (4.14) of $\alpha_2(\rho, z)$, by (3.26) and (3.27), one has

$$\alpha_3(\vartheta, \xi) = 4\xi + O(\xi^{\frac{3}{2}}), \quad \partial_\xi \alpha_3(\vartheta, \xi) = 4 + O(\xi^{\frac{1}{2}}) \quad \text{as } \xi \rightarrow 0, \xi > 0, \quad (3.32)$$

uniformly in $\vartheta \in \mathbb{T}$. Note that $\alpha_3(\vartheta, \xi)$ is not analytic around $\xi = 0$, namely $\alpha_3(\vartheta, \xi)$, as a function of ξ , is not a power series of ξ centered at zero; in fact, (3.32) is deduced from the analyticity of $\alpha_2(\rho, z)$ (and of its partial derivatives $\partial_\rho \alpha_2(\rho, z)$, $\partial_z \alpha_2(\rho, z)$) around $(1, 0)$, and then from the evaluation at $(\rho, z) = (1 + \sqrt{2\xi} \sin \vartheta, \sqrt{2\xi} \cos \vartheta)$. See Sect. 4.3 for more details.

Moreover, $\alpha_3(\vartheta, 0) = \alpha_2(1, 0) = 0$, and, by (3.32),

$$\partial_\xi \alpha_3(\vartheta, 0) = \lim_{\xi \rightarrow 0^+} \frac{\alpha_3(\vartheta, \xi) - \alpha_3(\vartheta, 0)}{\xi} = 4 = \lim_{\xi \rightarrow 0^+} \partial_\xi \alpha_3(\vartheta, \xi). \quad (3.33)$$

Thus α_3 is C^1 in $\mathbb{T} \times [0, \xi_3]$. Taking ξ_3 smaller (i.e., δ smaller) if necessary, we have $\partial_\xi \alpha_3(\vartheta, \xi) > 0$ for all $(\vartheta, \xi) \in \mathbb{T} \times [0, \xi_3]$. Hence the function $\xi \mapsto \alpha_3(\vartheta, \xi)$ is strictly increasing on $[0, \xi_3]$. Moreover $\alpha_3(\vartheta, 0) = 0$ and, by (3.29), $\alpha_3(\vartheta, \xi_3/4) > 4\tau$ for all $\vartheta \in \mathbb{T}$.

As a consequence, for every $c \in [0, 4\tau]$, for every $\vartheta \in \mathbb{T}$, there exists a unique $\xi \in [0, \xi_3/4]$ such that $\alpha_3(\vartheta, \xi) = c$. We denote $\gamma_c(\vartheta)$ the unique solution ξ of the equation $\alpha_3(\vartheta, \xi) = c$. Thus,

$$\alpha_3(\vartheta, \gamma_c(\vartheta)) = c \quad \forall (\vartheta, c) \in \mathbb{T} \times [0, 4\tau]. \quad (3.34)$$

Moreover, $\gamma_0(\vartheta) = 0$. Since α_3 is analytic in $(\vartheta, \xi) \in \mathbb{T} \times (0, \xi_3)$, by the implicit function theorem, the function $\gamma_c(\vartheta)$ is analytic in $(\vartheta, c) \in \mathbb{T} \times (0, 4\tau)$. The behavior of $\gamma_c(\vartheta)$ around $c = 0$ is studied in Sect. 4.3. From (3.34),

$$(\partial_\vartheta \alpha_3)(\vartheta, \gamma_c(\vartheta)) + (\partial_\xi \alpha_3)(\vartheta, \gamma_c(\vartheta)) \partial_\vartheta \gamma_c(\vartheta) = 0, \quad (3.35)$$

$$(\partial_\xi \alpha_3)(\vartheta, \gamma_c(\vartheta)) \partial_c \gamma_c(\vartheta) = 1 \quad (3.36)$$

for all $(\vartheta, c) \in \mathbb{T} \times (0, 4\tau)$. Since $\partial_\xi \alpha_3(\vartheta, \xi)$ is positive for all $(\vartheta, \xi) \in \mathbb{T} \times [0, \xi_3]$, by (3.36) it follows that

$$\partial_c \gamma_c(\vartheta) > 0 \quad \forall (\vartheta, c) \in \mathbb{T} \times (0, 4\tau). \quad (3.37)$$

Thus, for all $c \in [0, 4\tau]$, the level set $\mathcal{P}_{3,c}$ is globally described as the graph

$$\mathcal{P}_{3,c} = \{(\vartheta, \varphi, \xi) \in \mathbb{T} \times \mathbb{T} \times [0, \xi_3] : \xi = \gamma_c(\vartheta)\} = \{(\vartheta, \varphi, \gamma_c(\vartheta)) : (\vartheta, \varphi) \in \mathbb{T}^2\}. \quad (3.38)$$

3.5. Integration of the System in the Radial–Vertical Plane

We consider the Cauchy problem of system (3.24) with initial datum $(\vartheta_0, \varphi_0, \xi_0) \in \mathcal{S}_3^*$ at the initial time $t = 0$. The components ϑ_0 and ξ_0 of the initial datum determine the level $c = \alpha_3(\vartheta_0, \xi_0)$; then the solution $(\vartheta(t), \varphi(t), \xi(t))$ of the Cauchy problem is in the level set $\mathcal{P}_{3,c}$ for all $t \in \mathbb{R}$, and, by (3.38), $\xi(t) = \gamma_c(\vartheta(t))$ for all $t \in \mathbb{R}$.

The first and third equations of system (3.24) are the Hamiltonian system (3.30), i.e.,

$$\dot{\vartheta} = \chi(\alpha_3(\vartheta, \xi)) \partial_\xi \alpha_3(\vartheta, \xi), \quad \dot{\xi} = -\chi(\alpha_3(\vartheta, \xi)) \partial_\vartheta \alpha_3(\vartheta, \xi). \quad (3.39)$$

Since $\xi(t) = \gamma_c(\vartheta(t))$, (3.39) becomes

$$\dot{\vartheta} = \chi(c) (\partial_\xi \alpha_3)(\vartheta, \gamma_c(\vartheta)), \quad \partial_\vartheta \gamma_c(\vartheta) \dot{\vartheta} = -\chi(c) (\partial_\vartheta \alpha_3)(\vartheta, \gamma_c(\vartheta)). \quad (3.40)$$

By (3.35), the second equation in (3.40) is equal to the first equation in (3.40) multiplied by $\partial_{\vartheta}\gamma_c(\vartheta)$, and therefore system (3.40) is equivalent to its first equation alone. Moreover, by (3.36), the first equation in (3.40) is also

$$\dot{\vartheta} = \frac{\chi(c)}{\partial_c \gamma_c(\vartheta)}, \quad (3.41)$$

where the denominator is nonzero by (3.36) or (3.37). Equation (3.41) with initial datum $\vartheta(0) = \vartheta_0 \in \mathbb{T}$ is an autonomous Cauchy problem for the function $\vartheta(t)$ taking values in $\mathbb{T}$, and it can be integrated by basic calculus.

The function $\gamma_c(\vartheta)$ and its partial derivative $\partial_c \gamma_c(\vartheta)$ are defined for $\vartheta \in \mathbb{T}$, and hence they can also be considered as functions defined for $\vartheta \in \mathbb{R}$ that are 2π -periodic in ϑ . Thus, we first solve (3.41) considered as an equation for a function $\vartheta^r(t)$ taking values in $\mathbb{R}$; then the equivalence class of $\vartheta^r(t)$ mod 2π will be a function $\vartheta(t)$ taking values in $\mathbb{T}$ and solving (3.41).

For all $c \in (0, 4\tau)$, we define

$$F_c : \mathbb{R} \rightarrow \mathbb{R}, \quad F_c(\vartheta) := \int_0^{\vartheta} \partial_c \gamma_c(\sigma) d\sigma. \quad (3.42)$$

By (3.37), F_c is a diffeomorphism of $\mathbb{R}$. Let $\vartheta_0^r \in \mathbb{R}$ be a representative of the equivalence class $\vartheta_0 \in \mathbb{T}$. If $\vartheta^r : \mathbb{R} \rightarrow \mathbb{R}$, $t \mapsto \vartheta^r(t)$ solves (3.41) with $\vartheta^r(0) = \vartheta_0^r$, then

$$\frac{d}{dt}\{F_c(\vartheta^r(t))\} = F'_c(\vartheta^r(t))\dot{\vartheta}^r(t) = \chi(c), \quad F_c(\vartheta^r(t)) = F_c(\vartheta_0^r) + \chi(c)t, \quad (3.43)$$

and

$$\vartheta^r(t) = F_c^{-1}(F_c(\vartheta_0^r) + \chi(c)t). \quad (3.44)$$

Hence $\vartheta^r(t)$ in (3.44) is the unique solution of equation (3.41) with initial datum $\vartheta^r(0) = \vartheta_0^r$. Now let $\vartheta(t)$ be the equivalence class mod 2π of $\vartheta^r(t)$, i.e.,

$$\vartheta(t) := \{\vartheta^r(t) + 2k\pi : k \in \mathbb{Z}\}, \quad (3.45)$$

for all $t \in \mathbb{R}$. Then the function $\vartheta(t)$ in (3.45) is a function of time taking values in $\mathbb{T}$, and it solves (3.41) with $\vartheta(0) = \vartheta_0$.

3.6. Rotation Period in the Radial-Vertical Plane

We show that the function $\vartheta(t)$ in (3.45) is periodic, and we calculate its period. We begin with observing that the function F_c defined in (3.42) satisfies

$$F_c(2\pi + \vartheta) = F_c(2\pi) + F_c(\vartheta) \quad \forall \vartheta \in \mathbb{R}. \quad (3.46)$$

Proof of (3.46). Consider the definition (3.42), split the integral over $[0, 2\pi + \vartheta]$ into the sum of (i) the integral over $[0, 2\pi]$ and (ii) the integral over $[2\pi, 2\pi + \vartheta]$; then (i) = $F(2\pi)$ by definition, while (ii) = $F(\vartheta)$ because the function $\partial_c \gamma(\sigma, c)$ is 2π -periodic in σ and therefore (ii) is equal to the integral over $[0, \vartheta]$. $\square$

Suppose that $\chi(c)$ is nonzero. Then the function $\vartheta^r(t)$ defined in (3.44) satisfies

$$\vartheta^r(t + T_c) = \vartheta^r(t) + 2\pi \quad \forall t \in \mathbb{R}, \quad T_c := \frac{F_c(2\pi)}{\chi(c)}. \quad (3.47)$$

Proof of (3.47). Applying (3.43) twice, one has

$$F_c(\vartheta^r(t + T_c)) = F_c(\vartheta_0^r) + \chi(c)(t + T_c) = F_c(\vartheta^r(t)) + \chi(c)T_c.$$

By the definition of T_c in (3.47) and by identity (3.46),

$$F_c(\vartheta^r(t)) + \chi(c)T_c = F_c(\vartheta^r(t)) + F_c(2\pi) = F_c(\vartheta^r(t) + 2\pi).$$

Hence $F_c(\vartheta^r(t + T_c)) = F_c(\vartheta^r(t) + 2\pi)$, and, since F_c is invertible, we obtain (3.47). $\square$

From (3.47) it follows that $\vartheta(t)$ defined in (3.45) is periodic in time with period T_c .

3.7. Angle-Action Variables in the Radial-Vertical Plane

The Hamiltonian system (3.30) has one degree of freedom, and therefore, by the classical theory of Hamiltonian systems, it is completely integrable both in the sense that the differential equations can be solved by quadrature (i.e., they can be transformed into a problem of finding primitives of functions), and in the sense that they admit “angle-action variables” (this is the one-dimensional, simplest case of the Liouville-Arnold theorem). System (3.30) has been integrated in Sect. 3.5; now we construct its angle-action variables.

Given a point $(\vartheta^*, \varphi^*, \xi^*) \in \mathcal{S}_3^*$, we calculate its level $c = \alpha_3(\vartheta^*, \xi^*)$, which is in the interval $(0, 4\tau)$, and we consider the Cauchy problem

$$\dot{\vartheta} = \frac{1}{\partial_c \gamma_c(\vartheta)}, \quad \vartheta(0) = 0. \quad (3.48)$$

The first equation in (3.48) is equation (3.41) in which $\chi(c)$ is replaced by 1. Hence, following Sect. 3.5 line by line with 1 instead of $\chi(c)$, the $\mathbb{R}$ -valued solution of (3.48) is given by (3.44) in which $\chi(c)$ is replaced by 1; note that the functions $\alpha_3(\vartheta, \xi)$, $\gamma_c(\vartheta)$, $F_c(\vartheta)$ are all independent of $\chi(c)$. We indicate $\vartheta_c^r(t)$ the $\mathbb{R}$ -valued solution of (3.48). Thus, since $F_c(0) = 0$,

$$\vartheta_c^r(t) = F_c^{-1}(t) \quad \forall t \in \mathbb{R}. \quad (3.49)$$

Moreover, following Sect. 3.6 with 1 instead of $\chi(c)$, one has

$$\vartheta_c^r(t + T_c^*) = \vartheta_c^r(t) + 2\pi \quad \forall t \in \mathbb{R}, \quad T_c^* := F_c(2\pi). \quad (3.50)$$

We fix the representative ϑ_1^* of the class $\vartheta^* \in \mathbb{T}$ such that $\vartheta_1^* \in [0, 2\pi)$, and we take the unique real number $s_1^* \in [0, T_c^*)$ such that $\vartheta_c^r(s_1^*) = \vartheta_1^*$, i.e., by (3.49), $s_1^* = F_c(\vartheta_1^*)$. Then we define $\sigma_1^* = s_1^* 2\pi / T_c^*$, and we note that $\sigma_1^* \in [0, 2\pi)$ because $s_1^* \in [0, T_c^*)$. By the definition of σ_1^*, s_1^*, T_c^* , one has $\sigma_1^* = f_c(\vartheta_1^*)$, where

$$f_c : \mathbb{R} \rightarrow \mathbb{R}, \quad f_c(\vartheta) := \frac{F_c(\vartheta) 2\pi}{F_c(2\pi)} \quad \forall \vartheta \in \mathbb{R}. \quad (3.51)$$

Also, $\vartheta_1^* = \vartheta_c^r(s_1^*)$ and $s_1^* = \sigma_1^* T_c^* / 2\pi$, whence $\vartheta_1^* = g_c(\sigma_1^*)$, where

$$g_c : \mathbb{R} \rightarrow \mathbb{R}, \quad g_c(\sigma) := \vartheta_c^r\left(\frac{\sigma T_c^*}{2\pi}\right) \quad \forall \sigma \in \mathbb{R}. \quad (3.52)$$

The function f_c is a diffeomorphism of $\mathbb{R}$ because it is a multiple of F_c , and it satisfies

$$f_c(\vartheta + 2\pi) = f_c(\vartheta) + 2\pi \quad \forall \vartheta \in \mathbb{R} \quad (3.53)$$

because F_c satisfies (3.46). The function g_c also satisfies

$$g_c(\sigma + 2\pi) = g_c(\sigma) + 2\pi \quad \forall \sigma \in \mathbb{R} \quad (3.54)$$

because ϑ_c^r satisfies (3.50). By (3.49), (3.51), (3.52), $g_c(f_c(\vartheta)) = \vartheta$ for all $\vartheta \in \mathbb{R}$ and $f_c(g_c(\sigma)) = \sigma$ for all $\sigma \in \mathbb{R}$, i.e., g_c is the inverse diffeomorphism of f_c .

By (3.53) and (3.54), f_c and g_c induce diffeomorphisms of the torus

$$f_c : \mathbb{T} \rightarrow \mathbb{T}, \quad g_c : \mathbb{T} \rightarrow \mathbb{T}, \quad g_c(f_c(\vartheta)) = \vartheta \quad \forall \vartheta \in \mathbb{T}, \quad f_c(g_c(\sigma)) = \sigma \quad \forall \sigma \in \mathbb{T}$$

(with a common little abuse, we use the same notation for the diffeomorphisms of $\mathbb{R}$ and for the corresponding induced diffeomorphism of $\mathbb{T}$). Now σ_1^* and ϑ_1^* are related by the identities $\vartheta_1^* = g_c(\sigma_1^*)$, $\sigma_1^* = f_c(\vartheta_1^*)$. By the definition of ϑ_1^* , $\vartheta^* \in \mathbb{T}$ is the equivalence class of ϑ_1^* mod 2π ; let $\sigma^* \in \mathbb{T}$ be the equivalence class of σ_1^* mod 2π . Then

$$\vartheta^* = g_c(\sigma^*), \quad \sigma^* = f_c(\vartheta^*), \quad \vartheta^*, \sigma^* \in \mathbb{T}.$$

By (3.34), $\xi^* = \gamma_c(\vartheta^*)$. Thus, we have expressed (ϑ^*, ξ^*) in terms of (σ^*, c) . From now on, we write σ instead of σ^* .

Now we introduce a variable I and a function h to express the level c in terms of I , i.e., $c = h(I)$; the function h is a diffeomorphism (to be determined) of some interval $(0, I^*)$ (to be determined) onto the interval $(0, 4\tau)$ to which c belongs. We consider the map

$$\Phi_4(\sigma, \varphi, I) := (g_c(\sigma), \varphi, \gamma_c(g_c(\sigma)))|_{c=h(I)} = (g_{h(I)}(\sigma), \varphi, \gamma_{h(I)}(g_{h(I)}(\sigma))), \quad (3.55)$$

defined on the set

$$\mathcal{S}_4^* := \{(\sigma, \varphi, I) : \sigma, \varphi \in \mathbb{T}, I \in (0, I^*)\} = \mathbb{T} \times \mathbb{T} \times (0, I^*), \quad (3.56)$$

and we want to calculate its Jacobian determinant. One has

$$\begin{aligned} \partial_\sigma \{g_c(\sigma)\} &= g'_c(\sigma), & \partial_I \{g_c(\sigma)\} &= \partial_c g_c(\sigma) h'(I), \\ \partial_\sigma \{\gamma_c(g_c(\sigma))\} &= \gamma'_c(g_c(\sigma)) g'_c(\sigma), \\ \partial_I \{\gamma_c(g_c(\sigma))\} &= \{(\partial_c \gamma_c)(g_c(\sigma)) + \gamma'_c(g_c(\sigma)) \partial_c g_c(\sigma)\} h'(I) \end{aligned}$$

where $c = h(I)$. Hence the Jacobian matrix is

$$D\Phi_4(\sigma, \varphi, I) = \begin{bmatrix} g'_c(\sigma) & 0 & \partial_c g_c(\sigma) h'(I) \\ 0 & 1 & 0 \\ \gamma'_c(g_c(\sigma)) g'_c(\sigma) & 0 & \{(\partial_c \gamma_c)(g_c(\sigma)) + \gamma'_c(g_c(\sigma)) \partial_c g_c(\sigma)\} h'(I) \end{bmatrix}$$

and its determinant is

$$\det D\Phi_4(\sigma, \varphi, I) = g'_c(\sigma) (\partial_c \gamma_c)(g_c(\sigma)) h'(I)$$

where $c = h(I)$. By (3.52) and (3.48),

$$g'_c(\sigma) = \frac{1}{(\partial_c \gamma_c)(g_c(\sigma))} \frac{T_c^*}{2\pi},$$

and $T_c^* = F_c(2\pi)$, see (3.50). Hence

$$\det D\Phi_4(\sigma, \varphi, I) = \frac{F_c(2\pi)}{2\pi} h'(I) \quad (3.57)$$

where $c = h(I)$. We want the determinant (3.57) to be $= 1$, so that the transformation Φ_4 is symplectic and the Hamiltonian structure is preserved. The average

$$h_1(c) := \frac{1}{2\pi} \int_0^{2\pi} \gamma_c(\vartheta) d\vartheta \quad (3.58)$$

is a strictly increasing function of $c \in [0, 4\tau]$ because its derivative $h'_1(c) = F_c(2\pi)/2\pi$ is positive for all $c \in (0, 4\tau)$ by (3.37) and (3.42). Since $\gamma_0(\vartheta) = 0$, one has $h_1(0) = 0$. We define

$$I^* := h_1(4\tau), \quad (3.59)$$

and we note that, by the definition of $\gamma_c(\vartheta)$ in Sect. 3.4, one has $0 < I^* < \xi_3/4$. Thus, the interval $[0, I^*]$ is the image $\{h_1(c) : c \in [0, 4\tau]\}$ of the interval $[0, 4\tau]$. We define $h : [0, I^*] \rightarrow [0, 4\tau]$ as the inverse of $h_1 : [0, 4\tau] \rightarrow [0, I^*]$. Hence $h(0) = 0$, $h(I^*) = 4\tau$,

$$h'(I) > 0 \quad \forall I \in (0, I^*), \quad (3.60)$$

and

$$\det D\Phi_4(\sigma, \varphi, I) = 1 \quad \forall (\sigma, \varphi, I) \in \mathcal{S}_4^*. \quad (3.61)$$

This concludes the definition of the map Φ_4 in (3.55), which is a symplectic diffeomorphism of $\mathcal{S}_4^*$ onto $\mathcal{S}_3^*$.

Now that the transformation Φ_4 has been defined, we use it to transform system (3.24). We consider the change of variables $\tilde{v} = \Phi_4(v)$, where $\tilde{v} = (\vartheta, \varphi, \xi) \in \mathcal{S}_3^*$, $v = (\sigma, \varphi, I) \in \mathcal{S}_4^*$. This means that

$\vartheta = g_{h(I)}(\sigma)$ and $\xi = \gamma_{h(I)}(g_{h(I)}(\sigma))$, that is, $\vartheta = g_c(\sigma)$ and $\xi = \gamma_c(g_c(\sigma))$ where $c = h(I)$. A function $\tilde{v}(t) = \Phi_4(v(t))$ solves (3.24) in $\mathcal{S}_3^*$ if and only if $v(t)$ solves

$$\dot{v}(t) = V_4(v(t)) \quad (3.62)$$

in $\mathcal{S}_4^*$, where

$$V_4(v) := (D\Phi_4(v))^{-1}V_3(\Phi_4(v)), \quad v = (\sigma, \varphi, I) \in \mathcal{S}_4^*. \quad (3.63)$$

The second row of the inverse matrix $(D\Phi_4(v))^{-1}$ is $(0, 1, 0)$, and therefore the second component of the vector field V_4 is simply the second component of V_3 (see (3.28)) evaluated at $\Phi_4(\sigma, \varphi, I)$, which is $\chi(h(I))Q(\sigma, I)$, where

$$Q(\sigma, I) := \frac{\sqrt{H(h(I))}}{(1 + \sqrt{2\gamma_{h(I)}(g_{h(I)}(\sigma)) \sin(g_{h(I)}(\sigma))})^2}. \quad (3.64)$$

Since the first and third equations of (3.62) are the Hamiltonian system (3.30) and Φ_4 is symplectic in the (σ, I) variables, the first and third components of the vector field V_4 are $\partial_I \mathcal{H}_4(\sigma, I)$ and $-\partial_\sigma \mathcal{H}_4(\sigma, I)$ respectively, where $\mathcal{H}_4 := \mathcal{H}_3 \circ \Phi_4$ (of course this can also be checked directly, without using the properties of the symplectic transformations). Now $\mathcal{H}_3$ is defined in (3.31), with Γ defined in (3.19). Hence

$$\mathcal{H}_4(\sigma, I) = \mathcal{H}_4(I) = \Gamma(h(I)), \quad \partial_I \mathcal{H}_4(I) = \chi(h(I))h'(I), \quad (3.65)$$

and system (3.62) is

$$\dot{\sigma} = \partial_I \mathcal{H}_4(I), \quad \dot{\varphi} = \chi(h(I))Q(\sigma, I), \quad \dot{I} = 0 \quad (3.66)$$

in $\mathcal{S}_4^*$, with $Q(\sigma, I)$ defined in (3.64). The action I is constant in time. The angle σ rotates with constant angular velocity $\partial_I \mathcal{H}_4(I) = \chi(h(I))h'(I)$.

3.8. Reduction to a Constant Rotation in the Tangential Direction

Now that the equations of the motion in the radial-vertical plane have been written in angle-action variables (σ, I) in (3.66), we want to obtain a similar simplification for the equation of the motion in the tangential direction $\dot{\varphi}$. Recalling the definition (3.56) of $\mathcal{S}_4^*$, we consider the diffeomorphism

$$\Phi_5 : \mathcal{S}_4^* \rightarrow \mathcal{S}_4^*, \quad \Phi_5(\sigma, \beta, I) = (\sigma, \beta + \eta(\sigma, I), I), \quad (3.67)$$

where $\eta : \mathbb{T} \times (0, I^*) \rightarrow \mathbb{R}$ is a function to be determined. We consider the change of variables $\tilde{v} = \Phi_5(v)$, where $\tilde{v} = (\sigma, \varphi, I) \in \mathcal{S}_4^*$, $v = (\sigma, \beta, I) \in \mathcal{S}_4^*$. A function $\tilde{v}(t) = \Phi_5(v(t))$ solves (3.62) in $\mathcal{S}_4^*$ if and only if $v(t)$ solves

$$\dot{v}(t) = V_5(v(t)) \quad (3.68)$$

in $\mathcal{S}_4^*$, where

$$V_5(v) := (D\Phi_5(v))^{-1}V_4(\Phi_5(v)), \quad v = (\sigma, \beta, I) \in \mathcal{S}_4^*.$$

The Jacobian matrix and its inverse are

$$D\Phi_5(v) = \begin{bmatrix} 1 & 0 & 0 \\ \partial_\sigma \eta & 1 & \partial_I \eta \\ 0 & 0 & 1 \end{bmatrix}, \quad (D\Phi_5(v))^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -\partial_\sigma \eta & 1 & -\partial_I \eta \\ 0 & 0 & 1 \end{bmatrix}. \quad (3.69)$$

By (3.65) and (3.66), the vector field V_4 in (3.63) is

$$V_4(\sigma, \varphi, I) = \chi(h(I)) (h'(I), Q(\sigma, I), 0).$$

Hence

$$V_5(\sigma, \beta, I) = \chi(h(I)) (h'(I), Q(\sigma, I) - \partial_\sigma \eta(\sigma, I)h'(I), 0). \quad (3.70)$$

We decompose $Q(\sigma, I)$ as the sum of its average in $\sigma \in \mathbb{T}$ and its zero-average remainder, that is,

$$Q(\sigma, I) = Q_0(I) + \tilde{Q}(\sigma, I), \quad Q_0(I) := \frac{1}{2\pi} \int_0^{2\pi} Q(\sigma, I) d\sigma, \quad \tilde{Q} := Q - Q_0, \quad (3.71)$$

and define

$$\eta(\sigma, I) := \frac{1}{h'(I)} \int_0^\sigma \tilde{Q}(s, I) ds. \quad (3.72)$$

Note that $h'(I)$ is nonzero by (3.60). The function η is 2π -periodic in σ because $\tilde{Q}$ is 2π -periodic in σ with zero average on $\mathbb{T}$. By (3.72) and (3.71),

$$\partial_\sigma \eta(\sigma, I) h'(I) = \tilde{Q}(\sigma, I), \quad Q(\sigma, I) - \partial_\sigma \eta(\sigma, I) h'(I) = Q_0(I).$$

Thus, the vector field V_5 depends only on the action variable I , and it is

$$V_5(\sigma, \beta, I) = V_5(I) = (\Omega_1(I), \Omega_2(I), 0),$$

where

$$\Omega_1(I) := \partial_I \mathcal{H}_4(I) = \chi(h(I)) h'(I), \quad \Omega_2(I) := \chi(h(I)) Q_0(I). \quad (3.73)$$

Then system (3.68) is

$$\dot{\sigma} = \Omega_1(I), \quad \dot{\beta} = \Omega_2(I), \quad \dot{I} = 0. \quad (3.74)$$

3.9. Ratio of the Two Rotation Periods

Let $(\sigma_0, \beta_0, I_0) \in \mathcal{S}_4^* = \mathbb{T}^2 \times (0, I^*)$. Let $\sigma_0^r \in \mathbb{R}$ be a representative of the equivalence class $\sigma_0 \in \mathbb{T}$, and let $\beta_0^r \in \mathbb{R}$ be a representative of $\beta_0 \in \mathbb{T}$. The solution of the Cauchy problem for (3.74) in $\mathbb{R}^2 \times (0, I^*)$ with initial data $(\sigma_0^r, \beta_0^r, I_0)$ — i.e., system (3.74) where the first two equations are considered as equations for functions $\sigma^r(t)$, $\beta^r(t)$ taking values in $\mathbb{R}$ with initial data σ_0^r, β_0^r — is

$$\sigma^r(t) = \sigma_0^r + \Omega_1(I_0)t, \quad \beta^r(t) = \beta_0^r + \Omega_2(I_0)t, \quad I(t) = I_0 \quad \forall t \in \mathbb{R}. \quad (3.75)$$

Let $\sigma(t), \beta(t)$ be the equivalence classes of $\sigma^r(t), \beta^r(t)$ mod 2π , i.e.,

$$\sigma(t) := \{\sigma^r(t) + 2k\pi : k \in \mathbb{Z}\}, \quad \beta(t) := \{\beta^r(t) + 2k\pi : k \in \mathbb{Z}\}. \quad (3.76)$$

Then $(\sigma(t), \beta(t), I(t))$ is a function of time, taking values in $\mathcal{S}_4^* = \mathbb{T}^2 \times (0, I^*)$, solving (3.74) with initial datum (σ_0, β_0, I_0) .

If $\chi(h(I_0))$ is zero, then both $\Omega_1(I_0)$ and $\Omega_2(I_0)$ are zero by (3.73), and the solution $(\sigma(t), \beta(t), I(t))$ is constant in time. If, instead, $\chi(h(I_0))$ is nonzero, then $\Omega_1(I_0)$ is also nonzero by (3.73) and (3.60), and the function $\sigma(t)$ is periodic in time, with frequency $\Omega_1(I_0)$ and period $T_1(I_0) := 2\pi/\Omega_1(I_0)$. By (3.73), and recalling that the determinant in (3.57) is $= 1$, the period $T_1(I_0)$ is equal to the period T_c defined in (3.47) with $c = h(I_0)$. If $\Omega_2(I_0)$ is nonzero, then the function $\beta(t)$ is also periodic in time, with frequency $\Omega_2(I_0)$ and period $T_2(I_0) := 2\pi/\Omega_2(I_0)$.

We observe that the factor $\chi(h(I))$ cancels out in the frequency ratio

$$\frac{\Omega_2(I)}{\Omega_1(I)} = \frac{Q_0(I)}{h'(I)}, \quad (3.77)$$

and $Q_0(I)/h'(I)$ is well-defined even if $\chi(h(I))$ is zero. Hence $Q_0(I)/h'(I)$ is well-defined for all $I \in (0, I^*)$. By (3.71) and (3.64),

$$\frac{Q_0(I)}{h'(I)} = \frac{\sqrt{H(h(I))}}{h'(I)2\pi} \int_0^{2\pi} \frac{1}{(1 + \sqrt{2\gamma_{h(I)}(g_{h(I)}(\sigma))} \sin(g_{h(I)}(\sigma)))^2} d\sigma.$$

We make the change of variable $g_{h(I)}(\sigma) = \vartheta$ in the integral, that is, $\sigma = f_c(\vartheta)$, where $c = h(I)$ and f_c is defined in (3.51). Then $d\sigma = f'_c(\vartheta)d\vartheta$, and, by (3.51) and (3.42), $f'_c(\vartheta) = \partial_c \gamma_c(\vartheta) 2\pi / F_c(2\pi)$. Hence

$$\int_0^{2\pi} \frac{1}{(1 + \sqrt{2\gamma_{h(I)}(g_{h(I)}(\sigma))} \sin(g_{h(I)}(\sigma)))^2} d\sigma = \frac{2\pi}{F_c(2\pi)} \int_0^{2\pi} \frac{\partial_c \gamma_c(\vartheta)}{[1 + \sqrt{2\gamma_c(\vartheta)} \sin \vartheta]^2} d\vartheta$$

where $c = h(I)$. Moreover $h'(I)F_c(2\pi) = 2\pi$ because $h'(I)F_c(2\pi)/2\pi$ is the determinant in (3.57), which is 1. Therefore

$$\frac{Q_0(I)}{h'(I)} = A(h(I)), \quad A(c) := \sqrt{H(c)} J(c), \quad J(c) := \frac{1}{2\pi} \int_0^{2\pi} \frac{\partial_c \gamma_c(\vartheta)}{[1 + \sqrt{2\gamma_c(\vartheta)} \sin \vartheta]^2} d\vartheta. \quad (3.78)$$

3.10. Motion of the Fluid Particles

We consider the composition

$$\Phi := \Phi_1 \circ \Phi_2 \circ \Phi_3 \circ \Phi_4 \circ \Phi_5 \quad (3.79)$$

of the transformations defined in (3.3), (3.11), (3.23), (3.55), (3.67). The map Φ is defined on $\mathcal{S}_4^* = \mathbb{T}^2 \times (0, I^*)$, its image $\Phi(\mathcal{S}_4^*)$ is the set $\mathcal{S}^*$ in (1.10), and Φ is a diffeomorphism of $\mathcal{S}_4^*$ onto $\mathcal{S}^*$. The image $(x, y, z) = \Phi(\sigma, \beta, I) \in \mathcal{S}^*$ of a point $(\sigma, \beta, I) \in \mathcal{S}_4^*$ is

$$x = \varrho(\sigma, I) \cos(\beta + \eta(\sigma, I)), \quad y = \varrho(\sigma, I) \sin(\beta + \eta(\sigma, I)), \quad z = \zeta(\sigma, I),$$

with

$$\varrho(\sigma, I) := 1 + \sqrt{2\gamma_c(g_c(\sigma))} \sin(g_c(\sigma)), \quad \zeta(\sigma, I) := \frac{\sqrt{2\gamma_c(g_c(\sigma))} \cos(g_c(\sigma))}{1 + \sqrt{2\gamma_c(g_c(\sigma))} \sin(g_c(\sigma))}, \quad (3.80)$$

where $c = h(I)$. The analytic regularity of the map Φ and its expansion around $I = 0$ are studied in Sect. 4.8.

By construction, a function $\tilde{v}(t) = \Phi(v(t))$ is the solution of the Cauchy problem (1.11) with initial datum $\tilde{v}_0 = \Phi(v_0) \in \mathcal{S}^*$ if and only if the function $v(t)$ is the solution of (3.68), i.e., (3.74), with initial datum $v_0 \in \mathcal{S}_4^*$. As a consequence, recalling (3.75), (3.76), the solution $\tilde{v}(t)$ of the Cauchy problem (1.11) with initial datum $\tilde{v}_0 = (x_0, y_0, z_0) = \Phi(\sigma_0, \beta_0, I_0)$ is the function

$$\tilde{v}(t) = (x(t), y(t), z(t)) = \Phi(\sigma(t), \beta(t), I(t)) = \Phi(\sigma_0 + \Omega_1(I_0)t, \beta_0 + \Omega_2(I_0)t, I_0). \quad (3.81)$$

The function in (3.81) has the form $\tilde{v}(t) = w(\Omega_1 t, \Omega_2 t)$ where $w : \mathbb{T}^2 \rightarrow \mathcal{S}^*$ is the function $w(\vartheta_1, \vartheta_2) = \Phi(\sigma_0 + \vartheta_1, \beta_0 + \vartheta_2, I_0)$ and $\Omega_i = \Omega_i(I_0)$, $i = 1, 2$. Hence $\tilde{v}(t)$ is quasi-periodic with frequency vector (Ω_1, Ω_2) if $\Omega_1 \neq 0$, the ratio Ω_2/Ω_1 is irrational, and the number of frequencies cannot be reduced, i.e., if $\tilde{v}(t)$ is not a periodic function.

Now suppose that Ω_1 is nonzero, that Ω_2/Ω_1 is irrational, and that $\tilde{v}(t)$ in (3.81) is periodic with a certain period $T > 0$. Then, by (3.81),

$$\begin{pmatrix} \sigma_0 + \Omega_1(t+T) \\ \beta_0 + \Omega_2(t+T) \\ I_0 \end{pmatrix} = \Phi^{-1}(\tilde{v}(t+T)) = \Phi^{-1}(\tilde{v}(t)) = \begin{pmatrix} \sigma_0 + \Omega_1 t \\ \beta_0 + \Omega_2 t \\ I_0 \end{pmatrix} \quad \forall t \in \mathbb{R}.$$

Hence $\sigma_0 + \Omega_1(t+T)$ and $\sigma_0 + \Omega_1 t$ are the same element of $\mathbb{T}$. This means that $\Omega_1 T = 2\pi n$ for some $n \in \mathbb{Z}$. Similarly, $\beta_0 + \Omega_2(t+T) = \beta_0 + \Omega_2 t$ in $\mathbb{T}$, and $\Omega_2 T = 2\pi m$ for some $m \in \mathbb{Z}$. Since Ω_1 is nonzero, n is also nonzero, and $\Omega_2/\Omega_1 = m/n$ is rational, a contradiction. This proves that, for Ω_1 nonzero and Ω_2/Ω_1 irrational, the function $\tilde{v}(t)$ in (3.81) is not periodic, and therefore it is quasi-periodic with frequency vector (Ω_1, Ω_2) .

3.11. The Pressure in Terms of the Action

The pressure and the action are related in the following way. By (1.4), (3.2) and (3.3), one has $P(\Phi_1(\rho, \varphi, z)) = p(\rho, z) = \frac{1}{4}\alpha(\rho, z)$. By (3.11), (3.15),

$$P(\Phi_1(\Phi_2(\rho, \varphi, z))) = P(\Phi_1(\rho, \varphi, z\rho^{-1})) = \frac{1}{4}\alpha(\rho, z\rho^{-1}) = \frac{1}{4}\alpha_2(\rho, z).$$

By (3.23) and (3.26),

$$P(\Phi_1(\Phi_2(\Phi_3(\vartheta, \varphi, \xi)))) = \frac{1}{4}\alpha_2(1 + \sqrt{2\xi} \sin \vartheta, \sqrt{2\xi} \cos \vartheta) = \frac{1}{4}\alpha_3(\vartheta, \xi).$$

By (3.34) and (3.55),

$$P(\Phi_1(\Phi_2(\Phi_3(\Phi_4(\sigma, \varphi, I)))) = \frac{1}{4}\alpha_3(g_c(\sigma), \gamma_c(g_c(\sigma)))|_{c=h(I)} = \frac{1}{4}c|_{c=h(I)} = \frac{1}{4}h(I).$$

By (3.67) and (3.79),

$$P(\Phi(\sigma, \beta, I)) = \frac{1}{4}h(I). \quad (3.82)$$

For notation convenience, we define

$$\mathcal{K}(I) := \frac{1}{4}h(I), \quad (3.83)$$

so that $P(\Phi(\sigma, \beta, I)) = \mathcal{K}(I)$.

As a consequence, given any $\ell \in (0, \tau)$, the level set $\mathcal{T}_\ell = \{(x, y, z) \in \mathcal{S}^* : P(x, y, z) = \ell\}$ of the pressure is the image

$$\mathcal{T}_\ell = \Phi(\mathbb{T}^2 \times \{I\}) \quad (3.84)$$

of the set $\mathbb{T}^2 \times \{I\} \subseteq \mathcal{S}_4^*$, where $I \in (0, I^*)$ satisfies $\mathcal{K}(I) = \ell$. Moreover, $\mathcal{K}(I^*) = \frac{1}{4}h(I^*) = \tau$, see (3.59) and the lines just after it.

3.12. Volume of the Region Enclosed by a Level Set

By the definition of Φ_2 in (3.11), for all $(\rho, \varphi, z) \in \mathcal{N}_2$ one has

$$D(\Phi_1 \circ \Phi_2)(\rho, \varphi, z) = (D\Phi_1)(\Phi_2(\rho, \varphi, z))D\Phi_2(\rho, \varphi, z) = (D\Phi_1)(\rho, \varphi, z\rho^{-1})D\Phi_2(\rho, \varphi, z).$$

By (3.5) and (3.14),

$$\begin{aligned} \det D\Phi_1(\rho, \varphi, z) &= \rho & \forall (\rho, \varphi, z) \in \mathcal{N}_1, \\ \det D\Phi_2(\rho, \varphi, z) &= \rho^{-1} & \forall (\rho, \varphi, z) \in \mathcal{N}_2, \end{aligned}$$

and therefore

$$\det D(\Phi_1 \circ \Phi_2)(\rho, \varphi, z) = 1 \quad \forall (\rho, \varphi, z) \in \mathcal{N}_2.$$

By (3.25), (3.61) and (3.69), one has

$$\begin{aligned} \det D\Phi_3(\vartheta, \varphi, \xi) &= 1 & \forall (\vartheta, \varphi, \xi) \in \mathcal{B}_3, \\ \det D\Phi_4(\sigma, \varphi, I) &= 1 & \forall (\sigma, \varphi, I) \in \mathcal{S}_4^*, \\ \det D\Phi_5(\sigma, \beta, I) &= 1 & \forall (\sigma, \beta, I) \in \mathcal{S}_4^*. \end{aligned}$$

Hence, by the definition (3.79) of Φ , using repeatedly the chain rule and the formula for the determinant of the product of matrices, we deduce that $\det D\Phi(\sigma, \beta, I) = 1$ for all $(\sigma, \beta, I) \in \mathcal{S}_4^*$. As a consequence, the map Φ is volume-preserving.

Now fix any $\ell_0 \in (0, \tau)$, and consider the set $E_{\ell_0} = \{(x, y, z) \in \mathcal{N} : P(x, y, z) < \ell_0\}$. Since the circle $\mathcal{C}$ in (3.1) has zero volume, the open sets E_{ℓ_0} and $E_{\ell_0} \setminus \mathcal{C}$ have the same volume. One has

$$E_{\ell_0} \setminus \mathcal{C} = \{(x, y, z) \in \mathcal{N} : 0 < P(x, y, z) < \ell_0\} \subseteq \mathcal{S}^*,$$

and $E_{\ell_0} \setminus \mathcal{C}$ is the union of the pressure level sets $\mathcal{T}_\ell$ over $\ell \in (0, \ell_0)$. Hence, by (3.84),

$$E_{\ell_0} \setminus \mathcal{C} = \Phi(\mathbb{T}^2 \times (0, I_0)),$$

where $I_0 \in (0, I^*)$ satisfies $\mathcal{K}(I_0) = \ell_0$. Since the Jacobian determinant of Φ is 1, one has

$$\int_{E_{\ell_0} \setminus \mathcal{C}} dx \, dy \, dz = \int_{\mathbb{T}^2 \times (0, I_0)} |\det D\Phi(\sigma, \beta, I)| \, d\sigma \, d\beta \, dI = \int_{\mathbb{T}^2 \times (0, I_0)} d\sigma \, d\beta \, dI = (2\pi)^2 I_0.$$

4. Taylor Expansions and Transversality

In this section we prove that the frequency $\Omega_1(I)$ and the ratio $\Omega_2(I)/\Omega_1(I)$, see (3.73) and (3.77), admit a Taylor expansion around $I = 0$ (which, in principle, is not obvious because of the square roots in the construction), and we calculate the first nonzero coefficient in their expansion after the constant term. This forces us to expand the function α to degree 6. As a consequence, we obtain that $\Omega_1(I)$ and $\Omega_2(I)/\Omega_1(I)$ really change as I changes, and they vary in a smooth, strictly monotonic way, passing across every value of an interval exactly one time. This can be geometrically viewed as a transversality property. We also calculate the expansion of $\Phi(\sigma, \beta, I)$ around $I = 0$.

4.1. Expansion of α

The function α is constructed in [10] by solving the PDE system (4.4), where the functions F and G are expressed in terms of a function ψ , see (4.2) and (4.3). The function ψ is defined in [10] as the solution of a degenerate ODE problem. In Section 2.1 of [10] one finds the Taylor expansion of ψ of order 5 around zero; here we only use its expansion of order 3, which is

$$\psi(s) = 1 - \frac{3}{4}s + \frac{9}{128}s^2 - \frac{21}{1024}s^3 + O(s^4) \quad \text{as } s \rightarrow 0. \quad (4.1)$$

In Section 2.1 of [10] the functions H, F, G are defined in terms of ψ as

$$H(s) = 6s \left(\frac{1}{\psi'(s)} + 2\psi(s) \right), \quad F(x, s) = -2x\psi(s) + 2x^3, \quad (4.2)$$

$$G(x, s) = 12x^2s - F^2(x, s) - H(s). \quad (4.3)$$

In Lemma 3 of [10] the analytic function $\alpha(x, y)$ is defined as the unique solution of the system

$$\partial_x \alpha(x, y) = F(x, \alpha(x, y)), \quad (\partial_y \alpha(x, y))^2 = G(x, \alpha(x, y)) \quad (4.4)$$

in a neighborhood of $(x, y) = (1, 0)$ such that $\alpha(1, 0) = 0$, with $\partial_y \alpha$ not identically zero. In Remark 2 of [10] it is observed that α is even in y , i.e., $\alpha(x, y) = \alpha(x, -y)$. The coefficients of the monomials of degree 2 and 3 in the Taylor series

$$\alpha(x, y) = 2(x-1)^2 + 2y^2 + 3(x-1)^3 + 3(x-1)y^2 + \sum_{\substack{k, j \geq 0 \\ k+2j \geq 4}} \alpha_{k, 2j} (x-1)^k y^{2j} \quad (4.5)$$

are given in Remark 4 of [10]; here we want to calculate the coefficients of the monomials of degree 4, 5 and 6. Monomials with odd exponent $2j+1$ are not present in (4.5) because α is even in y . One has

$$\begin{aligned} \partial_x \alpha(x, y) &= 4(x-1) + 9(x-1)^2 + 3y^2 + 4\alpha_{40}(x-1)^3 + 2\alpha_{22}(x-1)y^2 \\ &\quad + 5\alpha_{50}(x-1)^4 + 3\alpha_{32}(x-1)^2 y^2 + \alpha_{14}y^4 + 6\alpha_{60}(x-1)^5 \\ &\quad + 4\alpha_{42}(x-1)^3 y^2 + 2\alpha_{24}(x-1)y^4 + O_6, \end{aligned} \quad (4.6)$$

where O_n denotes terms with homogeneity $\geq n$ in $(x-1, y)$. Since $\alpha(x, y) = O_2$, by (4.1) and (4.5) one has

$$\begin{aligned}\psi(\alpha(x, y)) &= 1 - \frac{3}{4}\alpha(x, y) + \frac{9}{128}\alpha^2(x, y) + O_6 \\ &= 1 - \frac{3}{2}(x-1)^2 - \frac{3}{2}y^2 - \frac{9}{4}(x-1)^3 - \frac{9}{4}(x-1)y^2 + \left(\frac{9}{32} - \frac{3}{4}\alpha_{40}\right)(x-1)^4 \\ &\quad + \left(\frac{9}{16} - \frac{3}{4}\alpha_{22}\right)(x-1)^2y^2 + \left(\frac{9}{32} - \frac{3}{4}\alpha_{04}\right)y^4 + \left(\frac{27}{32} - \frac{3}{4}\alpha_{50}\right)(x-1)^5 \\ &\quad + \left(\frac{27}{16} - \frac{3}{4}\alpha_{32}\right)(x-1)^3y^2 + \left(\frac{27}{32} - \frac{3}{4}\alpha_{14}\right)(x-1)y^4 + O_6.\end{aligned}$$

Expanding x and x^3 around $x = 1$, and recalling the definition (4.2) of F , we calculate

$$\begin{aligned}F(x, \alpha(x, y)) &= -2x\psi(\alpha(x, y)) + 2x^3 \\ &= -2\psi(\alpha(x, y)) - 2(x-1)\psi(\alpha(x, y)) + 2 + 6(x-1) + 6(x-1)^2 + 2(x-1)^3 \\ &= 4(x-1) + 9(x-1)^2 + 3y^2 + \frac{19}{2}(x-1)^3 + \frac{15}{2}(x-1)y^2 \\ &\quad + \left(\frac{63}{16} + \frac{3}{2}\alpha_{40}\right)(x-1)^4 + \left(\frac{27}{8} + \frac{3}{2}\alpha_{22}\right)(x-1)^2y^2 + \left(-\frac{9}{16} + \frac{3}{2}\alpha_{04}\right)y^4 \\ &\quad + \left(-\frac{9}{4} + \frac{3}{2}\alpha_{50} + \frac{3}{2}\alpha_{40}\right)(x-1)^5 + \left(\frac{9}{4} + \frac{3}{2}\alpha_{32} + \frac{3}{2}\alpha_{22}\right)(x-1)^3y^2 \\ &\quad + \left(\frac{9}{8} + \frac{3}{2}\alpha_{14} + \frac{3}{2}\alpha_{04}\right)(x-1)y^4 + O_6.\end{aligned}\tag{4.7}$$

From (4.6), (4.7) and the identity of each monomial in the differential equation $\partial_x \alpha = F(x, \alpha)$ we get

$$\begin{aligned}\alpha_{40} &= \frac{19}{8}, & \alpha_{22} &= \frac{15}{4}, & \alpha_{50} &= \frac{3}{2}, & \alpha_{32} &= 3, & \alpha_{14} &= -\frac{9}{16} + \frac{3}{4}\alpha_{04}, \\ \alpha_{60} &= \frac{19}{32}, & \alpha_{42} &= \frac{99}{32}, & \alpha_{24} &= \frac{9}{16} + \frac{3}{4}(\alpha_{14} + \alpha_{04}).\end{aligned}$$

To find the value of α_{04} and α_{06} , we consider $\partial_y \alpha(x, y)$ and $G(x, \alpha(x, y))$ at $x = 1$ and we expand them around $y = 0$. By (4.5), since α is even in y , one has

$$\alpha(1, y) = 2y^2 + \alpha_{04}y^4 + \alpha_{06}y^6 + O(y^8),\tag{4.8}$$

whence

$$\begin{aligned}\partial_y \alpha(1, y) &= 4y + 4\alpha_{04}y^3 + 6\alpha_{06}y^5 + O(y^7), \\ (\partial_y \alpha(1, y))^2 &= 16y^2 + 32\alpha_{04}y^4 + (48\alpha_{06} + 16\alpha_{04}^2)y^6 + O(y^8).\end{aligned}\tag{4.9}$$

By (4.7),

$$\begin{aligned}F(1, \alpha(1, y)) &= 3y^2 + \left(-\frac{9}{16} + \frac{3}{2}\alpha_{04}\right)y^4 + O(y^6), \\ F^2(1, \alpha(1, y)) &= 9y^4 + \left(-\frac{27}{8} + 9\alpha_{04}\right)y^6 + O(y^8).\end{aligned}$$

By (4.1) and (4.2),

$$H(s) = 4s - \frac{21}{2}s^2 + \frac{39}{32}s^3 + O(s^4)\tag{4.10}$$

(it is to obtain (4.10) that we use the coefficient of s^3 in (4.1)). Therefore, by (4.8),

$$H(\alpha(1, y)) = 8y^2 + (4\alpha_{04} - 42)y^4 + \left(4\alpha_{06} - 42\alpha_{04} + \frac{39}{4}\right)y^6 + O(y^8).$$

Hence

$$\begin{aligned} G(1, \alpha(1, y)) &= 12\alpha(1, y) - F^2(1, \alpha(1, y)) - H(\alpha(1, y)) \\ &= 16y^2 + (8\alpha_{04} + 33)y^4 + \left(8\alpha_{06} + 33\alpha_{04} + \frac{105}{8}\right)y^6 + O(y^8). \end{aligned} \quad (4.11)$$

By (4.9), (4.11) and the identity of each monomial in the differential equation $(\partial_y \alpha)^2 = G(x, \alpha)$ at $x = 1$ we get $\alpha_{04} = 11/8$ and $\alpha_{06} = 113/160$. Hence $\alpha_{14} = 15/32$, $\alpha_{24} = 249/128$, and

$$\begin{aligned} \alpha(x, y) &= 2(x-1)^2 + 2y^2 + 3(x-1)^3 + 3(x-1)y^2 + \frac{19}{8}(x-1)^4 + \frac{15}{4}(x-1)^2y^2 \\ &\quad + \frac{11}{8}y^4 + \frac{3}{2}(x-1)^5 + 3(x-1)^3y^2 + \frac{15}{32}(x-1)y^4 + \frac{19}{32}(x-1)^6 \\ &\quad + \frac{99}{32}(x-1)^4y^2 + \frac{249}{128}(x-1)^2y^4 + \frac{113}{160}y^6 + O_7. \end{aligned} \quad (4.12)$$

4.2. Expansion of α_2

The function $\alpha_2(\rho, z) = \alpha(\rho, z/\rho)$ defined in (3.15) is even in z , it is analytic around $(\rho, z) = (1, 0)$, and its Taylor series

$$\alpha_2(\rho, z) = \sum_{\substack{k, j \geq 0 \\ k+2j \geq 2}} (\alpha_2)_{k, 2j} (\rho-1)^k y^{2j} \quad (4.13)$$

matches with the expansion in (4.12) in which $x = \rho$ and $y = z/\rho$. We expand

$$\begin{aligned} \rho^{-2} &= 1 - 2(\rho-1) + 3(\rho-1)^2 - 4(\rho-1)^3 + 5(\rho-1)^4 + O((\rho-1)^5), \\ \rho^{-4} &= 1 - 4(\rho-1) + 10(\rho-1)^2 + O((\rho-1)^3), \end{aligned}$$

and we obtain

$$\begin{aligned} \alpha_2(\rho, z) &= 2(\rho-1)^2 + 2z^2 + 3(\rho-1)^3 - (\rho-1)z^2 + \frac{19}{8}(\rho-1)^4 + \frac{15}{4}(\rho-1)^2y^2 \\ &\quad + \frac{11}{8}z^4 + \frac{3}{2}(\rho-1)^5 - \frac{7}{2}(\rho-1)^3z^2 - \frac{161}{32}(\rho-1)z^4 + \frac{19}{32}(\rho-1)^6 \\ &\quad + \frac{203}{32}(\rho-1)^4z^2 + \frac{1529}{128}(\rho-1)^2z^4 + \frac{113}{160}z^6 + O_7. \end{aligned} \quad (4.14)$$

4.3. Expansion of $\gamma_c(\vartheta)$

Recalling the definition (3.20) of $\mathcal{B}_2$ and the first inclusion in (3.21), the function $\alpha_2(\rho, z)$ in (3.15) is defined and analytic in the disc $(\rho-1)^2 + z^2 < \delta_2^2$. We define

$$\phi : \mathbb{T} \times (-\delta_2, \delta_2) \rightarrow \mathbb{R}^2, \quad \phi(\vartheta, r) := (1 + r \sin \vartheta, r \cos \vartheta), \quad \tilde{\alpha}_3(\vartheta, r) := \alpha_2(\phi(\vartheta, r)). \quad (4.15)$$

The function $\tilde{\alpha}_3$ is well-posed and analytic in $(\vartheta, r) \in \mathbb{T} \times (-\delta_2, \delta_2)$, and, by (4.13), it is the power series

$$\tilde{\alpha}_3(\vartheta, r) = \sum_{n=2}^{\infty} P_n(\vartheta) r^n, \quad P_n(\vartheta) = \sum_{\substack{k, j \geq 0 \\ k+2j=n}} (\alpha_2)_{k, 2j} (\sin \vartheta)^k (\cos \vartheta)^{2j}. \quad (4.16)$$

All $P_n(\vartheta)$ are trigonometric polynomials, and $P_n(-\vartheta) = (-1)^n P_n(\vartheta)$, i.e., P_n is even for n even and P_n is odd for n odd, because $(-1)^k = (-1)^n$ for $k+2j=n$. Hence

$$\tilde{\alpha}_3(-\vartheta, -r) = \tilde{\alpha}_3(\vartheta, r) \quad \forall (\vartheta, r) \in \mathbb{T} \times (-\delta_2, \delta_2), \quad (4.17)$$

namely $\tilde{\alpha}_3$ is an even function of the pair (ϑ, r) . In fact, (4.17) is the symmetry property $\alpha_2(\rho, -z) = \alpha_2(\rho, z)$ expressed in terms of the function $\tilde{\alpha}_3(\vartheta, r)$. By (4.14),

$$\begin{aligned} P_2(\vartheta) &= 2, \\ P_3(\vartheta) &= 3\sin^3 \vartheta - \sin \vartheta \cos^2 \vartheta, \\ P_4(\vartheta) &= \frac{19}{8}\sin^4 \vartheta + \frac{15}{4}\sin^2 \vartheta \cos^2 \vartheta + \frac{11}{8}\cos^4 \vartheta, \\ P_5(\vartheta) &= \frac{3}{2}\sin^5 \vartheta - \frac{7}{2}\sin^3 \vartheta \cos^2 \vartheta - \frac{161}{32}\sin \vartheta \cos^4 \vartheta, \\ P_6(\vartheta) &= \frac{19}{32}\sin^6 \vartheta + \frac{203}{32}\sin^4 \vartheta \cos^2 \vartheta + \frac{1529}{128}\sin^2 \vartheta \cos^4 \vartheta + \frac{113}{160}\cos^6 \vartheta. \end{aligned} \quad (4.18)$$

Hence

$$P_3(\vartheta) = 2\sin(\vartheta) - \sin(3\vartheta), \quad (4.19)$$

$$P_4(\vartheta) = \frac{15}{8} - \frac{1}{2}\cos(2\vartheta), \quad (4.20)$$

$$P_5(\vartheta) = -\frac{33}{256}\sin(\vartheta) - \frac{835}{512}\sin(3\vartheta) + (P_5)_5 \sin(5\vartheta), \quad (4.21)$$

$$P_6(\vartheta) = \frac{3173}{2048} + (P_6)_2 \cos(2\vartheta) + (P_6)_4 \cos(4\vartheta) + (P_6)_6 \cos(6\vartheta). \quad (4.22)$$

The Fourier coefficients $(P_5)_5$ in (4.21) and $(P_6)_k$, $k = 2, 4, 6$, in (4.22) are not involved in the calculations we are going to make, and therefore we avoid to calculate their numerical value.

When $\tilde{\alpha}_3(\vartheta, r)$ is evaluated at $r = \sqrt{2\xi}$, we obtain the function $\alpha_3(\vartheta, \xi)$ defined in (3.26), i.e.,

$$\alpha_3(\vartheta, \xi) = \tilde{\alpha}_3(\vartheta, \sqrt{2\xi}). \quad (4.23)$$

The function $\gamma_c(\vartheta)$ is defined in (3.34) as the unique solution ξ of the equation $\alpha_3(\vartheta, \xi) = c$. Because of the square root in the construction, $\gamma_c(\vartheta)$, as a function of c , is not analytic around $c = 0$, i.e., it is not a power series of the form $\sum Q_n(\vartheta)c^n$ for some analytic functions $Q_n(\vartheta)$. However, $\gamma_c(\vartheta)$ is a power series of the form $\sum Q_n(\vartheta)c^{n/2}$, namely there exists a function $\tilde{\gamma}(\vartheta, \mu)$, analytic around $\mu = 0$, such that $\gamma_c(\vartheta)$ is $\tilde{\gamma}(\vartheta, \mu)$ evaluated at $\mu = \sqrt{c}$. To prove it, we use the implicit function theorem for analytic functions, taking into account the degeneracy of the problem.

We define

$$\mathcal{F}(\vartheta, \mu, w) := \mu^{-2}\tilde{\alpha}_3(\vartheta, \mu w) - 1 \quad \text{if } \mu \neq 0; \quad \mathcal{F}(\vartheta, 0, w) := 2w^2 - 1. \quad (4.24)$$

The function $\mathcal{F}$ in (4.24) is well-defined and analytic in $\mathbb{T} \times (-\mu_0, \mu_0) \times (-w_0, w_0)$, for some $\mu_0, w_0 > 0$ small enough, and, by (4.16),

$$\mathcal{F}(\vartheta, \mu, w) = \sum_{n=2}^{\infty} P_n(\vartheta)w^n \mu^{n-2} - 1 = 2w^2 - 1 + \sum_{n=3}^{\infty} P_n(\vartheta)w^n \mu^{n-2}.$$

Moreover, by (4.17) and (4.24),

$$\mathcal{F}(-\vartheta, -\mu, w) = \mathcal{F}(\vartheta, \mu, w) \quad \forall (\vartheta, \mu, w) \in \mathbb{T} \times (-\mu_0, \mu_0) \times (-w_0, w_0). \quad (4.25)$$

For every $\vartheta \in \mathbb{T}$ one has

$$\mathcal{F}(\vartheta, 0, 2^{-\frac{1}{2}}) = 0, \quad \partial_w \mathcal{F}(\vartheta, 0, 2^{-\frac{1}{2}}) = 4 \cdot 2^{-\frac{1}{2}} \neq 0.$$

Hence there exist two constants μ_1, w_1 , with $0 < \mu_1 \leq \mu_0$, $0 < w_1 \leq w_0$, and a function $w(\vartheta, \mu)$, defined and analytic in $\mathbb{T} \times (-\mu_1, \mu_1)$, taking values in $(-w_1, w_1)$, such that

$$w(\vartheta, 0) = 2^{-\frac{1}{2}} \quad \forall \vartheta \in \mathbb{T}, \quad \mathcal{F}(\vartheta, \mu, w(\vartheta, \mu)) = 0 \quad \forall (\vartheta, \mu) \in \mathbb{T} \times (-\mu_1, \mu_1), \quad (4.26)$$

and such that if a point $(\vartheta, \mu, a) \in \mathbb{T} \times (-\mu_1, \mu_1) \times (-w_1, w_1)$ is a zero of $\mathcal{F}$, then $a = w(\vartheta, \mu)$.

By (4.26) and (4.25), for all (ϑ, μ) one has

$$0 = \mathcal{F}(-\vartheta, -\mu, w(-\vartheta, -\mu)) = \mathcal{F}(\vartheta, \mu, w(-\vartheta, -\mu)).$$

Hence the point $(\vartheta, \mu, w(-\vartheta, -\mu)) \in \mathbb{T} \times (-\mu_1, \mu_1) \times (-w_1, w_1)$ is a zero of $\mathcal{F}$, and therefore it belongs to the graph of the implicit function, i.e.,

$$w(-\vartheta, -\mu) = w(\vartheta, \mu) \quad \forall (\vartheta, \mu) \in \mathbb{T} \times (-\mu_1, \mu_1). \quad (4.27)$$

From the second identity in (4.26) and formula (4.16) it follows that $w(\vartheta, r)$ is the power series

$$w(\vartheta, \mu) = \sum_{n=0}^{\infty} W_n(\vartheta) \mu^n, \quad (4.28)$$

where the functions $W_n(\vartheta)$ are determined by the identity $\sum_{n=2}^{\infty} P_n(\vartheta) w^n(\vartheta, \mu) \mu^{n-2} = 1$, i.e., $W_0(\vartheta) = 2^{-\frac{1}{2}}$ and $W_n(\vartheta)$ are trigonometric polynomials recursively determined by the system

$$\sum_{\substack{n \geq 2, j \geq 0, \\ n-2+j=m}} \sum_{\substack{k_1, \dots, k_n \geq 0, \\ k_1 + \dots + k_n = j}} P_n(\vartheta) W_{k_1}(\vartheta) W_{k_2}(\vartheta) \cdots W_{k_n}(\vartheta) = 0 \quad \forall m \geq 1. \quad (4.29)$$

By (4.27) and (4.28), $W_n(-\vartheta) = (-1)^n W_n(\vartheta)$, i.e., W_n is even for n even and W_n is odd for n odd. We will make use of equations (4.29) for $m = 1, 2, 3, 4$, which are

$$2P_2 W_0 W_1 + P_3 W_0^3 = 0, \quad (4.30)$$

$$P_2(2W_0 W_2 + W_1^2) + 3P_3 W_0^2 W_1 + P_4 W_0^4 = 0, \quad (4.31)$$

$$P_2(2W_0 W_3 + 2W_1 W_2) + P_3(3W_0^2 W_2 + 3W_0 W_1^2) + 4P_4 W_0^3 W_1 + P_5 W_0^5 = 0, \quad (4.32)$$

$$P_2(2W_0 W_4 + 2W_1 W_3 + W_2^2) + P_3(3W_0^2 W_3 + 6W_0 W_1 W_2 + W_1^3) + P_4(4W_0^3 W_2 + 6W_0^2 W_1^2) + 5P_5 W_0^4 W_1 + P_6 W_0^6 = 0. \quad (4.33)$$

By the definition (4.24) of $\mathcal{F}$, the second identity in (4.26) implies that

$$\tilde{\alpha}_3(\vartheta, \mu w(\vartheta, \mu)) = \mu^2 \quad \forall (\vartheta, \mu) \in \mathbb{T} \times (-\mu_1, \mu_1) \quad (4.34)$$

with $\mu \neq 0$. Identity (4.34) also holds for $\mu = 0$ because $\tilde{\alpha}_3(\vartheta, 0) = \alpha_2(\phi(\vartheta, 0)) = \alpha_2(1, 0) = 0$.

By the first identity in (4.26), taking μ_1 smaller if necessary, one has $w(\vartheta, \mu) > 0$ for all $(\vartheta, \mu) \in \mathbb{T} \times (-\mu_1, \mu_1)$. Given any $c \in [0, \mu_1^2)$, there exists a unique $\mu \in [0, \mu_1)$ such that $c = \mu^2$, that is, $\mu = \sqrt{c}$. Also, $\mu w(\vartheta, \mu) \geq 0$, and there exists a unique $\xi \geq 0$ such that $\mu w(\vartheta, \mu) = \sqrt{2\xi}$, that is, $\xi = \frac{1}{2} \mu^2 w^2(\vartheta, \mu)$. As a consequence, by (4.23) and (4.34),

$$\alpha_3(\vartheta, \xi) = \tilde{\alpha}_3(\vartheta, \sqrt{2\xi}) = \tilde{\alpha}_3(\vartheta, \mu w(\vartheta, \mu)) = \mu^2 = c. \quad (4.35)$$

Hence $\xi = \gamma_c(\vartheta)$, and therefore $\gamma_c(\vartheta) = \frac{1}{2} \mu^2 w^2(\vartheta, \mu)$ where $\mu = \sqrt{c}$. In other words, we have proved that

$$\gamma_c(\vartheta) = \tilde{\gamma}(\vartheta, \sqrt{c}) \quad \forall (\vartheta, c) \in \mathbb{T} \times [0, \mu_1^2), \quad (4.36)$$

where $\tilde{\gamma}$ is the analytic function

$$\tilde{\gamma}(\vartheta, \mu) := \frac{1}{2} \mu^2 w^2(\vartheta, \mu). \quad (4.37)$$

By (4.27),

$$\tilde{\gamma}(-\vartheta, -\mu) = \tilde{\gamma}(\vartheta, \mu) \quad \forall (\vartheta, \mu) \in \mathbb{T} \times (-\mu_1, \mu_1). \quad (4.38)$$

By (4.37) and (4.28),

$$\tilde{\gamma}(\vartheta, \mu) = \sum_{n=2}^{\infty} Q_n(\vartheta) \mu^n, \quad Q_n(\vartheta) := \frac{1}{2} \sum_{\substack{k, j \geq 0 \\ k+j+2=n}} W_k(\vartheta) W_j(\vartheta). \quad (4.39)$$

By (4.38) and (4.39), one has $Q_n(-\vartheta) = (-1)^n Q_n(\vartheta)$.

4.4. Expansion of the Average of $\gamma_c(\vartheta)$ and $\partial_c \gamma_c(\vartheta)$

We study the average of $\tilde{\gamma}(\vartheta, \mu)$, $\gamma_c(\vartheta)$ and $\partial_c \gamma_c(\vartheta)$ over $\vartheta \in [0, 2\pi]$. To shorten the notation, given any 2π -periodic function $f(\vartheta)$, we denote $\langle f \rangle$ its average over the period, i.e.,

$$\langle f \rangle := \frac{1}{2\pi} \int_0^{2\pi} f(\vartheta) d\vartheta.$$

For n odd, the trigonometric polynomial $Q_n(\vartheta)$ in (4.39) is 2π -periodic and odd, and therefore $\langle Q_n \rangle = 0$. As a consequence, by (4.39),

$$\frac{1}{2\pi} \int_0^{2\pi} \tilde{\gamma}(\vartheta, \mu) d\vartheta = \sum_{k=1}^{\infty} \langle Q_{2k} \rangle \mu^{2k}. \quad (4.40)$$

By (3.58), (4.36) and (4.40), one has

$$h_1(c) = \frac{1}{2\pi} \int_0^{2\pi} \gamma_c(\vartheta) d\vartheta = \frac{1}{2\pi} \int_0^{2\pi} \tilde{\gamma}(\vartheta, \sqrt{c}) d\vartheta = \sum_{k=1}^{\infty} \langle Q_{2k} \rangle c^k. \quad (4.41)$$

Hence $h_1(c)$ is analytic around $c = 0$, in the sense that $h_1(c)$, which is defined for $c \in [0, \mu_1^2)$, coincides in $[0, \mu_1^2)$ with the power series in (4.41), which is a function defined for $c \in (-\mu_1^2, \mu_1^2)$ and analytic in that interval. Note that the average of $\gamma_c(\vartheta)$ is analytic around $c = 0$ even if the function $\gamma_c(\vartheta)$ itself is not analytic in c around $c = 0$.

Now we calculate the averages $\langle Q_n \rangle$ for $n = 2, 4, 6$. By (4.39),

$$Q_2 = \frac{1}{2} W_0^2 = \frac{1}{4}, \quad Q_4 = \frac{1}{2} (2W_0 W_2 + W_1^2), \quad Q_6 = \frac{1}{2} (2W_0 W_4 + 2W_1 W_3 + W_2^2). \quad (4.42)$$

Hence $\langle Q_2 \rangle = 1/4$. Since $P_2 = 2$ and $W_0 = 2^{-\frac{1}{2}}$, from (4.30) we get

$$W_1 = -\frac{1}{8} P_3. \quad (4.43)$$

Using (4.31) to substitute $(2W_0 W_2 + W_1^2)$, and using also (4.43), we calculate

$$Q_4 = -\frac{1}{4} (3P_3 W_0^2 W_1 + P_4 W_0^4) = \frac{1}{64} (3P_3^2 - 4P_4).$$

By (4.19) and (4.20), we obtain

$$\langle Q_4 \rangle = 0. \quad (4.44)$$

Using (4.33) to substitute $(2W_0 W_4 + 2W_1 W_3 + W_2^2)$, and also (4.43), we calculate

$$\begin{aligned} Q_6 &= -\frac{1}{4} [P_3 (3W_0^2 W_3 + 6W_0 W_1 W_2 + W_1^3) + P_4 (4W_0^3 W_2 + 6W_0^2 W_1^2) + 5P_5 W_0^4 W_1 + P_6 W_0^6] \\ &= -\frac{3}{8} P_3 W_3 + \frac{3\sqrt{2}}{32} P_3^2 W_2 + \frac{1}{2048} P_3^4 - \frac{\sqrt{2}}{4} P_4 W_2 - \frac{3}{256} P_3^2 P_4 + \frac{5}{128} P_3 P_5 - \frac{1}{32} P_6. \end{aligned} \quad (4.45)$$

From (4.31) we obtain

$$W_2 = \frac{5\sqrt{2}}{128} P_3^2 - \frac{\sqrt{2}}{16} P_4 = \frac{\sqrt{2}}{256} (-5 - 32 \cos(2\vartheta) + 20 \cos(4\vartheta) - 5 \cos(6\vartheta)) \quad (4.46)$$

and, from (4.32) and (4.43),

$$W_3 = \frac{P_3 P_4}{16} - \frac{P_3 W_2}{2\sqrt{2}} - \frac{3P_3^3}{256} - \frac{P_5}{16} = \frac{225}{4096} \sin(\vartheta) + \frac{1251}{8192} \sin(3\vartheta) + \sum_{k=5,7,9} (W_3)_k \sin(k\vartheta). \quad (4.47)$$

The Fourier coefficients $(W_3)_k$, $k = 5, 7, 9$, are not involved in the calculation of $\langle Q_6 \rangle$, and therefore we do not calculate them. By (4.19), ..., (4.22), (4.46), (4.47), we calculate

$$\begin{aligned} \langle P_3 W_3 \rangle &= -\frac{351}{16384}, & \langle P_3^2 W_2 \rangle &= \frac{291\sqrt{2}}{1024}, & \langle P_3^4 \rangle &= \frac{131}{8}, & \langle P_4 W_2 \rangle &= -\frac{11\sqrt{2}}{2048}, \\ \langle P_3^2 P_4 \rangle &= \frac{91}{16}, & \langle P_3 P_5 \rangle &= \frac{703}{1024}, & \langle P_6 \rangle &= \frac{3173}{2048}. \end{aligned}$$

Hence, integrating (4.45), we obtain

$$\langle Q_6 \rangle = -\frac{1065}{65536} \quad (4.48)$$

(where $65536 = 2^{16}$), and, by (4.41),

$$h_1(c) = \frac{1}{2\pi} \int_0^{2\pi} \gamma_c(\vartheta) d\vartheta = \frac{c}{4} + \langle Q_6 \rangle c^3 + O(c^4). \quad (4.49)$$

Since $\langle Q_6 \rangle$ in (4.48) is nonzero, $h_1(c)$ is a nonlinear, analytic function of c . Its derivative is

$$h'_1(c) = \frac{1}{2\pi} \int_0^{2\pi} \partial_c \gamma_c(\vartheta) d\vartheta = \frac{1}{4} + 3\langle Q_6 \rangle c^2 + O(c^3), \quad (4.50)$$

which is a nonconstant, analytic function of c .

4.5. Expansion of $J(c)$

The average $J(c)$ is defined in (3.78). By (4.36), the partial derivative $\partial_c \gamma_c(\vartheta)$ satisfies

$$\partial_c \gamma_c(\vartheta) = \nu(\vartheta, \sqrt{c}), \quad (4.51)$$

where $\nu(\vartheta, \mu)$ is the analytic function

$$\nu(\vartheta, \mu) := \frac{\partial_\mu \tilde{\gamma}(\vartheta, \mu)}{2\mu} = \sum_{n=2}^{\infty} Q_n(\vartheta) \frac{n}{2} \mu^{n-2}, \quad (4.52)$$

with $Q_n(\vartheta)$ defined in (4.39). By (4.38), one has $\nu(-\vartheta, -\mu) = \nu(\vartheta, \mu)$. Moreover

$$\nu(\vartheta, \mu) = Q_2 + \frac{3}{2} Q_3(\vartheta) \mu + 2Q_4(\vartheta) \mu^2 + O(\mu^3), \quad (4.53)$$

and $Q_2 = 1/4$ by (4.42), $Q_3 = W_0 W_1$ by (4.39), $\langle Q_3 \rangle = 0$ because Q_3 is odd, and $\langle Q_4 \rangle = 0$ by (4.44). Regarding the denominator in the definition of $J(c)$, one has $\sqrt{2\gamma_c(\vartheta)} = \sqrt{2\xi} = \sqrt{c} w(\vartheta, \sqrt{c})$ by construction (see (4.35) and the lines preceding it), and, by (4.28), (4.43), (4.19),

$$\begin{aligned} (1 + \mu w(\vartheta, \mu) \sin \vartheta)^{-2} &= 1 - 2\mu w(\vartheta, \mu) \sin \vartheta + 3\mu^2 w^2(\vartheta, \mu) \sin^2 \vartheta + O(\mu^3) \\ &= 1 - 2W_0 \sin(\vartheta) \mu + (3W_0^2 \sin^2 \vartheta - 2W_1(\vartheta) \sin \vartheta) \mu^2 + O(\mu^3) \\ &= 1 - \sqrt{2} \sin(\vartheta) \mu + \left(1 - \frac{9}{8} \cos(2\vartheta) + \frac{1}{8} \cos(4\vartheta)\right) \mu^2 + O(\mu^3). \end{aligned} \quad (4.54)$$

Taking a smaller μ_1 if necessary, the function

$$m(\vartheta, \mu) := \frac{\nu(\vartheta, \mu)}{[1 + \mu w(\vartheta, \mu) \sin \vartheta]^2} \quad (4.55)$$

is defined and analytic in $\mathbb{T} \times (-\mu_1, \mu_1)$, it satisfies

$$m(-\vartheta, -\mu) = m(\vartheta, \mu) \quad \forall (\vartheta, \mu) \in \mathbb{T} \times (-\mu_1, \mu_1), \quad (4.56)$$

and it has expansion $m(\vartheta, \mu) = \sum_{n=0}^{\infty} M_n(\vartheta) \mu^n$ for some trigonometric polynomials $M_n(\vartheta)$. From (4.56) it follows that $M_n(-\vartheta) = (-1)^n M_n(\vartheta)$. Therefore $\langle M_n \rangle = 0$ for n odd, and

$$\frac{1}{2\pi} \int_0^{2\pi} m(\vartheta, \mu) d\vartheta = \sum_{k=0}^{\infty} \langle M_{2k} \rangle \mu^{2k}.$$

By (4.53) and (4.54) one has

$$M_0 = \frac{1}{4}, \quad M_2(\vartheta) = 2Q_4(\vartheta) - \frac{3}{2}W_1(\vartheta) \sin \vartheta + \frac{1}{4} - \frac{9}{32} \cos(2\vartheta) + \frac{1}{32} \cos(4\vartheta).$$

By (4.43) and (4.19), the average of $W_1(\vartheta) \sin \vartheta$ is $-1/8$. Hence $\langle M_2 \rangle = 7/16$, and

$$\begin{aligned} J(c) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\partial_c \gamma_c(\vartheta)}{[1 + \sqrt{2\gamma_c(\vartheta)} \sin \vartheta]^2} d\vartheta \\ &= \frac{1}{2\pi} \int_0^{2\pi} m(\vartheta, \sqrt{c}) d\vartheta = \sum_{k=0}^{\infty} \langle M_{2k} \rangle c^k = \frac{1}{4} + \frac{7}{16}c + O(c^2). \end{aligned} \quad (4.57)$$

Thus, the average $J(c)$ is an analytic function of c around $c = 0$ (even if the integrand function $m(\vartheta, \sqrt{c})$ is not). Moreover, taking μ_1 smaller if necessary, $J(c)$ is strictly increasing in $[0, \mu_1^2]$.

4.6. Expansion of $h(I)$ and $\Omega_1(I)$

We have already proved that $h_1(c)$ in (3.58) is analytic around $c = 0$, with expansion (4.49). Hence its inverse function $h(I) = h_1^{-1}(I)$ is also analytic around $I = 0$, and it satisfies

$$h(I) = 4I - 256\langle Q_6 \rangle I^3 + O(I^4), \quad h'(I) = 4 - 768\langle Q_6 \rangle I^2 + O(I^3). \quad (4.58)$$

The frequency $\Omega_1(I)$ is defined in (3.73), and it is the product of the C^∞ cut-off function $\chi(h(I))$ times the analytic function $h'(I)$ in (4.58).

4.7. Expansion of the Frequency Ratio $\Omega_2(I)/\Omega_1(I)$

By (3.77) and (3.78), the frequency ratio $\Omega_2(I)/\Omega_1(I)$ coincides with the function

$$A(h(I)) = \sqrt{H(h(I))} J(h(I)).$$

By its definition in [10], the function $H(c)$ is analytic around $c = 0$. Hence the composition $H(h(I))$ is analytic around $I = 0$, and, by (4.10) and (4.58),

$$H(h(I)) = 16I - 168I^2 + O(I^3).$$

We write its square root as the product

$$\sqrt{H(h(I))} = \sqrt{I} B(I), \quad (4.59)$$

where

$$B(I) := 4 \left(\frac{H(h(I))}{16I} \right)^{\frac{1}{2}} = 4 \left(1 - \frac{21}{2}I + O(I^2) \right)^{\frac{1}{2}} = 4 - 21I + O(I^2). \quad (4.60)$$

Since the function $x \mapsto \sqrt{1+x}$ is analytic around $x = 0$, the function $B(I)$ is analytic around $I = 0$. The function $J(h(I))$ is also analytic around $I = 0$, and, by (4.57) and (4.58),

$$J(h(I)) = \frac{1}{4} + \frac{7}{4}I + O(I^2). \quad (4.61)$$

Hence

$$A(h(I)) = \sqrt{I} \mathcal{R}(I), \quad \mathcal{R}(I) := B(I)J(h(I)) = 1 + \frac{7}{4}I + O(I^2),$$

and the function $\mathcal{R}(I)$ is analytic around $I = 0$. Taking a smaller I^* if necessary, both $\mathcal{R}(I)$ and $A(h(I))$ are strictly increasing functions of $I \in [0, I^*)$.

4.8. Expansion of $\Phi(\sigma, \beta, I)$

Since $Q_2 = 1/4$, by (4.36), (4.39), (4.51), (4.52) one has

$$\gamma_c(\vartheta) = \frac{c}{4} + O(c^{\frac{3}{2}}), \quad \partial_c \gamma_c(\vartheta) = \frac{1}{4} + O(c^{\frac{1}{2}}).$$

Hence, by (3.42), $F_c(\vartheta) = \frac{\vartheta}{4} + O(c^{\frac{1}{2}})$ and therefore, by (3.51), $f_c(\vartheta) = \vartheta + O(c^{\frac{1}{2}})$. As a consequence, g_c in (3.52), being the inverse of f_c , satisfies

$$g_c(\vartheta) = \vartheta + O(c^{\frac{1}{2}}).$$

By (4.58), $h(I) = 4I + O(I^3)$ and $h'(I) = 4 + O(I^2)$. Hence, at $c = h(I)$, one has

$$\gamma_c(\vartheta) = I + O(I^{\frac{3}{2}}), \quad g_c(\vartheta) = \vartheta + O(I^{\frac{1}{2}}),$$

and therefore the functions $\varrho(\sigma, I), \zeta(\sigma, I)$ in (3.80) have expansion

$$\varrho(\sigma, I) = \sqrt{2I} \sin(\sigma) + O(I), \quad \zeta(\sigma, I) = \sqrt{2I} \cos(\sigma) + O(I). \quad (4.62)$$

By (4.59) and (4.60),

$$\sqrt{H(h(I))} = 4\sqrt{I} + O(I^{\frac{3}{2}}). \quad (4.63)$$

By (4.62) and (4.63), the function $Q(\sigma, I)$ in (3.64) satisfies

$$Q(\sigma, I) = 4I^{\frac{1}{2}} + O(I).$$

Hence $Q_0, \tilde{Q}, \eta$ in (3.71), (3.72) satisfy

$$Q_0(I) = 4I^{\frac{1}{2}} + O(I), \quad \tilde{Q}(\sigma, I) = O(I), \quad \eta(\sigma, I) = O(I).$$

The map Φ_1 in (3.3) is analytic in $\mathcal{N}_1$ because ρ does not vanish in $\mathcal{N}_1$; the map Φ_2 in (3.11) is also analytic in $\mathcal{N}_2$ because ρ does not vanish in $\mathcal{N}_2$.

The map Φ_3 in (3.23) is analytic in $\mathcal{B}_3$ because ξ is positive in $\mathcal{B}_3$; however, Φ_3 is not analytic in ξ around $\xi = 0$. Nonetheless, Φ_3 can be obtained by evaluating at $r = \sqrt{2\xi}$ a map that is analytic around $r = 0$, exactly like $\alpha_3(\vartheta, \xi)$ in (4.23), like $\gamma_c(\vartheta)$ in (4.36), and like $\partial_c \gamma_c(\vartheta)$ in (4.51).

Similarly, both the map Φ_4 in (3.55) and the map Φ_5 in (3.67) are analytic in $\mathcal{S}_4^*$, they are not analytic in I around $I = 0$, but they can be obtained by evaluating at $\mu = \sqrt{I}$ some suitable maps that are analytic around $\mu = 0$, because $\gamma_c, F_c, f_c, g_c, \sqrt{H(c)}$ are all functions of this type.

As a consequence, the map Φ in (3.79) is analytic in $\mathcal{S}_4^*$, it is not analytic in I around $I = 0$, and it can be obtained by evaluating at $\mu = \sqrt{I}$ a map that is analytic around $\mu = 0$. Hence Φ admits a converging expansion in powers of $\sqrt{I}$ around $I = 0$.

4.9. Smallness Conditions

The parameter δ in the definition (3.1) of the set $\mathcal{N}$ is subject to the following smallness conditions. After (3.1) we have taken $\delta \in (0, 1)$ to obtain that $\mathcal{N}$ is an open neighborhood of the circle $\mathcal{C}$ in (3.1) with $\rho > 1 - \delta > 0$ in $\mathcal{N}$, and $\delta \leq r_0$ to obtain that the functions $\alpha(\sqrt{x^2 + y^2}, z)$ and $H(\alpha(\sqrt{x^2 + y^2}, z))$ are analytic in $(x, y, z) \in \mathcal{N}$, where r_0 is a universal constant given by the definition of α and H , i.e., by Gavrilov's construction in [10]. After (3.20) we have defined $\delta_2 = 2\delta/3$, and we have assumed $\delta \leq 1/2$ to obtain the inclusion (3.21). After (3.23) we have defined $\xi_3 = \delta_2^2/2 = 2\delta^2/9$. In (3.33) we have proved that $\partial_\xi \alpha_3(\vartheta, 0) = 4$ for all $\vartheta \in \mathbb{T}$, and that $\partial_\xi \alpha_3(\vartheta, \xi)$ is continuous in $\mathbb{T} \times [0, \xi_3]$ — in fact, in Sect. 4.3 we have proved more, because $\alpha_3(\vartheta, \xi) = \tilde{\alpha}_3(\vartheta, \sqrt{2\xi})$, see (4.23), and $\tilde{\alpha}_3(\vartheta, r)$ is the analytic function in (4.15), (4.16); hence $\partial_\xi \alpha_3(\vartheta, \xi) = \sum_{n=2}^{\infty} P_n(\vartheta) n(2\xi)^{(n-2)/2} = 4 + 3P_3(\vartheta)\sqrt{2\xi} + O(\xi)$. Hence, by continuity,

there exists a universal constant $\xi^* > 0$ such that $\partial_\xi \alpha_3(\vartheta, \xi) > 0$ in $\mathbb{T} \times [0, \xi^*)$. Therefore the condition on ξ_3 after (3.33) (where we say “Taking ξ_3 smaller if necessary”) is $\xi_3 \leq \xi^*$. In terms of δ , this means $\delta \leq 3\sqrt{\xi^*}/2$, which is a universal constant.

The constants w_0, μ_0 after the definition (4.24) of $\mathcal{F}$ are universal, and the constants w_1, μ_1 after the application of the implicit function theorem in (4.26) are universal too. By (4.26), $w(\vartheta, 0) > 0$, and therefore, by continuity, there exists a universal constant $\mu_1^* > 0$ such that $w(\vartheta, \mu) > 0$ for all $(\vartheta, \mu) \in \mathbb{T} \times (-\mu_1^*, \mu_1^*)$. Hence the condition on μ_1 after (4.34) (where we say “taking μ_1 smaller if necessary”) is $\mu_1 \leq \mu_1^*$. The same happens for the condition on μ_1 after (4.54), to obtain that the function $m(\vartheta, \mu)$ in (4.55) is well-defined and analytic in $\mathbb{T} \times (-\mu_1, \mu_1)$, and for the condition on μ_1 after (4.57), to obtain that $J(c)$ is strictly increasing in $[0, \mu_1^2]$. Thus, we fix μ_1 as the smallest of these three constants, and we obtain that μ_1 is a universal positive constant.

The parameter τ is related to μ_1 by the inequality $4\tau \leq \mu_1^2$, because $[0, 4\tau)$ is the interval where c varies, and the functions $\gamma_c(\vartheta)$, $\partial_c \gamma_c(\vartheta)$ and $J(c)$ are obtained by evaluating at $\mu = \sqrt{c}$ some functions of (ϑ, μ) that are well-defined, analytic and monotonic in $\mathbb{T} \times (-\mu_1, \mu_1)$, see (4.36), (4.51), (4.57). Thus, we want that, for all $c \in [0, 4\tau)$, the square root $\sqrt{c}$ belongs to the domain $(-\mu_1, \mu_1)$ of those analytic functions, and this is true if $4\tau \leq \mu_1^2$.

Regarding the parameter τ , there are two other conditions to consider. The first one is (1.9), which holds if τ is smaller than the infimum of the pressure P on the set $\mathcal{N} \setminus \mathcal{N}'$. Since $P(x, y, z) = p(\rho, z) = \frac{1}{4}\alpha(\rho, z)$ (see (3.2)), by the definition (3.1) of $\mathcal{N}$ and $\mathcal{N}'$ it follows that that infimum depends only on δ . The last condition for τ is after (4.61), where we say “Taking a smaller I^* if necessary”, to obtain that $\mathcal{R}(I)$ is strictly increasing in $[0, I^*)$. Since the function $\mathcal{R}$ does not depend on any parameter, this is a condition of the form $I^* \leq I_0$ for some universal constant $I_0 > 0$. Moreover, the invertible function h_1 expressing I^* in terms of τ in (3.59) is also independent on any parameter (see the definition of h_1 in (3.58) and its expansion in (4.49)), and therefore this condition for τ is satisfied for τ smaller than a universal constant.

Regarding ε , the only condition to consider is that $0 < \varepsilon \leq \tau/3$, see after (1.9).

In conclusion, the parameters δ, τ and ε must satisfy

$$0 < \delta \leq \delta_0, \quad 0 < \tau \leq \tau_0(\delta), \quad 0 < \varepsilon \leq \tau/3,$$

where δ_0 is a universal constant, and $\tau_0(\delta)$ depends only on δ . We fix $\delta = \delta_0$ and $\tau = \tau_0(\delta_0)$. Both δ_0 and $\tau_0(\delta_0)$ are universal constants. We rename $\tau_0 := \tau_0(\delta_0)$ and $\varepsilon_0 := \tau_0/3$. Since $\tau = \tau_0$, by (3.59), I^* is also a universal constant.

For notation convenience, we consider the function $\mathcal{K}(I) = \frac{1}{4}h(I)$ defined in (3.83). Hence, by (3.73) and (3.7),

$$\Omega_1(I) = \frac{1}{4}\omega\left(\frac{1}{4}h(I)\right)h'(I) = \omega(\mathcal{K}(I))\mathcal{K}'(I);$$

by (4.58) and (4.48),

$$\mathcal{K}(I) = \frac{1}{4}\left(4I + 256\frac{1065}{2^{16}}I^3 + O(I^4)\right) = I + \frac{1065}{1024}I^3 + O(I^4).$$

For all $\varepsilon \in (0, \varepsilon_0]$, the proof of Theorem 1.1 is complete.

Acknowledgements. Supported by the Project PRIN 2020XB3EFL *Hamiltonian and dispersive PDEs*. The author is grateful to the two anonymous referees for their interesting comments and their stimulating questions.

Funding Open access funding provided by Università degli Studi di Napoli Federico II within the CRUI-CARE Agreement.

Declarations

Conflict of interest The author states that there is no conflict of interest.

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(accepted: October 5, 2023)